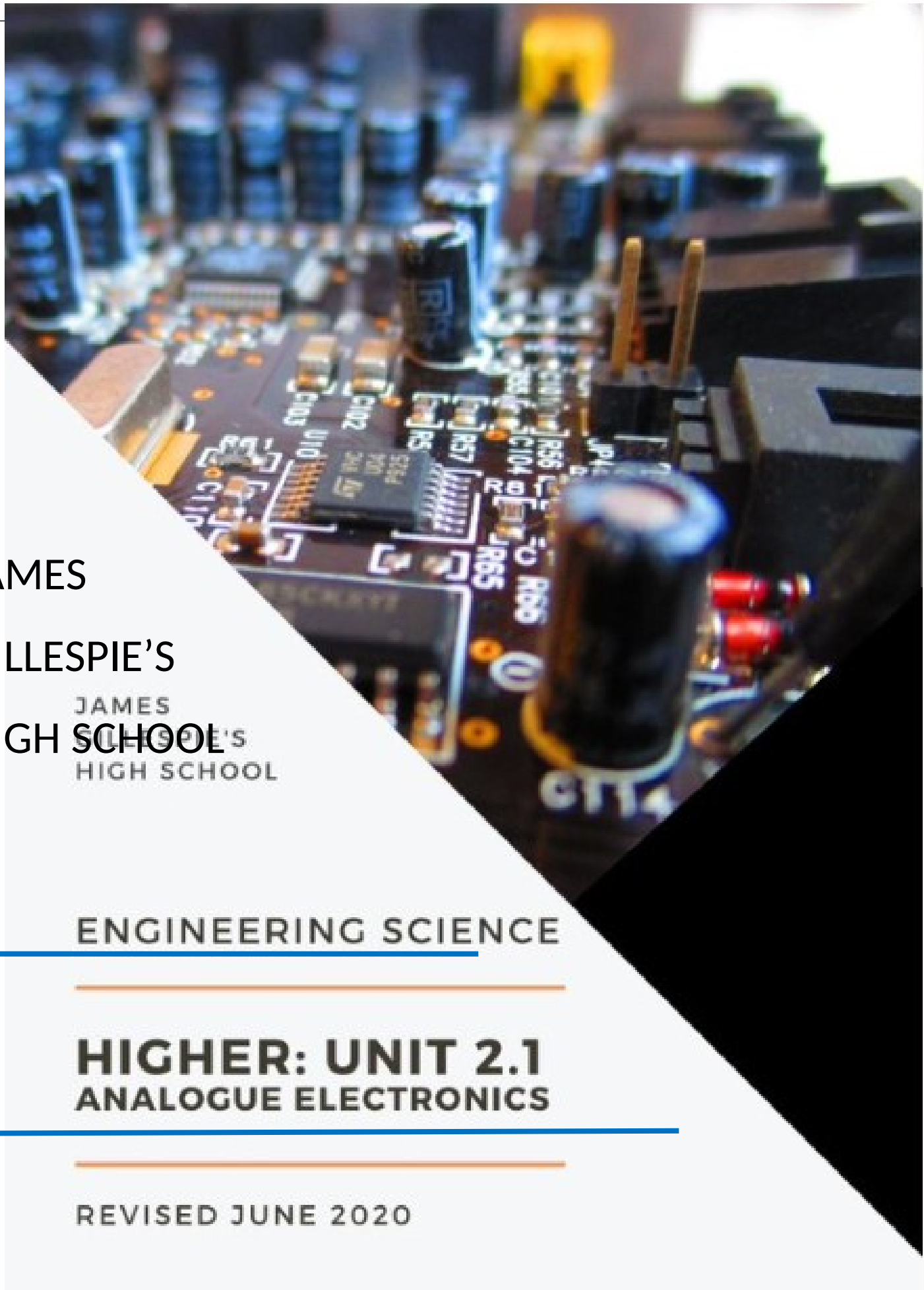


JAMES GILLESPIE'S HIGH SCHOOL

ENGINEERING SCIENCE

HIGHER: UNIT 2.1
ANALOGUE ELECTRONICS

REVISED JUNE 2020

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Learning Intentions

- To confidently make use of variable resistors, light and temperature sensors in voltage dividers
- To use input transducer characteristics, relevant to the above bullet, to design voltage dividers to meet a specification
- To understand the function and use of Op Amp configurations:
    - inverting
    - non-inverting
    - comparator
    - difference amplifier
    - summing amplifier
    - voltage followers
- To calculate relationship between input and output voltages for different Op Amp configurations
- To identify the function and purpose of BJTs
- To design a BJT circuit as a current amplifier
- To calculate the current gain (h_{FE}) of an npn transistor
- To identify function and purpose of MOSFETs
- To design a MOSFET (n-channel enhancement mode) circuit as a voltage-operated switch
- To compare BJT and MOSFET transistors in a given application

Success Criteria

Skills, knowledge and understanding for course assessment

In order to track your progress of learning, consider the above learning intentions and record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will review this record page after speaking to you about your self-evaluation.

☐ I can confidently use and evaluate input transducers and their characteristics

☐ I understand the function of 6 different Op-Amp configurations

☐ I can identify the function of BJTs and design a circuit with a BJT as a current amplifier

☐ I can identify the function and purpose of MOSFETs

☐ I can calculate h_{FE} for a npn transistor

☐ I can design a MOSFET circuit

☐ I can compare BJT and MOSFET transistors

Electronic circuits presumed knowledge

Below is an outline of your presumed basic knowledge of analogue electronics, as you enter Higher Engineering Science. If you do not have a good grasp of these topics, take time to ensure you refresh your understanding of:

- Series and parallel circuits
- Ohms law
- Kirchoff's laws for current and voltage
- Input and output transducers including:
    - Variable resistors
    - Thermistors
    - LDRs
- Voltage dividers
- Use of simulation software, Yenka

Input transducers

An input transducer is used to change a physical parameter into an electrical signal as the physical conditions change. In national 5 you came across temperature and light dependent resistors. They are always used in voltage divider circuits, so their output can be converted into a useful voltage change. This can either produce a direct measurement of the physical condition (temperature in °C) or allows something to happen at a predetermined level (e.g. switching ON the central heating at 20 °C).

The table gives some examples of common input transducers that you may have met before.

Physical condition monitored	Input transducer	Electrical property that changes
Temperature	Thermistor	Resistance
	Thermocouple	Voltage
	Platinum film	Resistance
Light	LDR	Resistance
	Selenium cell	Voltage

	Photo diode	Current/ resistance
Distance	Slide potentiometer	Resistance
Force	Strain gauge	Resistance
Angle	Rotary Potentiometer	Resistance

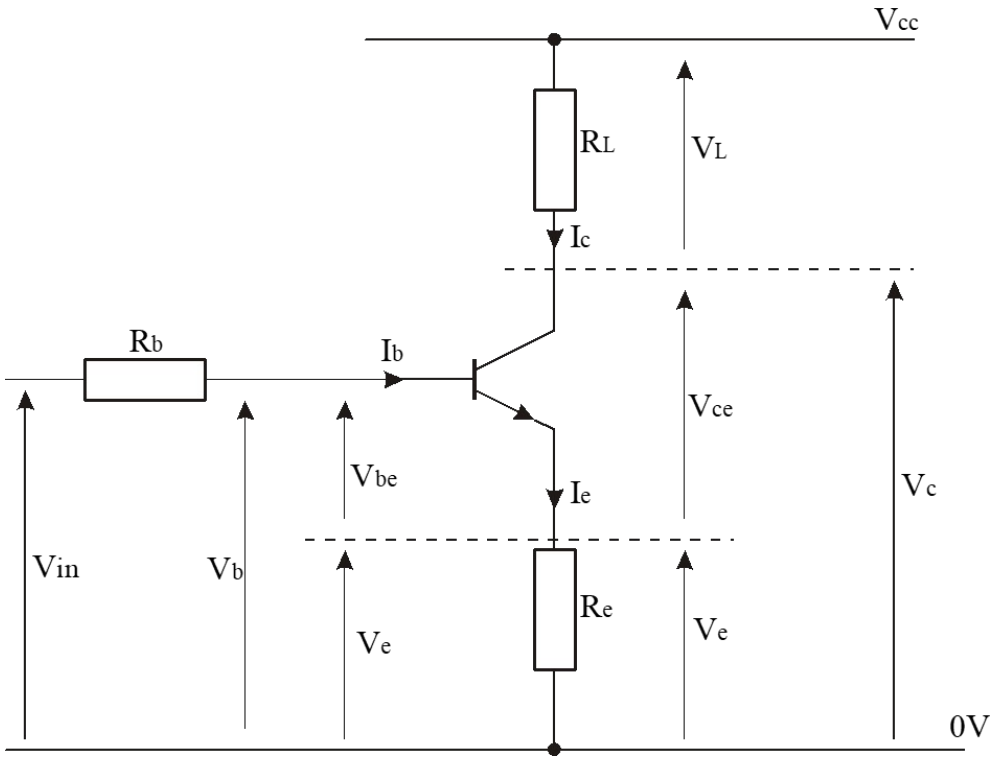
Transistor notation

The following may be useful as a reference page as you work through the questions.

I_c - Collector current
I_b - Base current
I_e - Emitter current
V_{CC} - Voltage of supply (relative to ground line)
V_b - Voltage at the base junction (relative to ground line)
V_e - Voltage at the emitter junction (relative to ground line)
V_{ce} - Voltage between the collector and emitter junction
V_{be} - Voltage between the base and emitter junction
V_L - Voltage over the load resistor
Av - Current gain
h_{FE} - Current gain

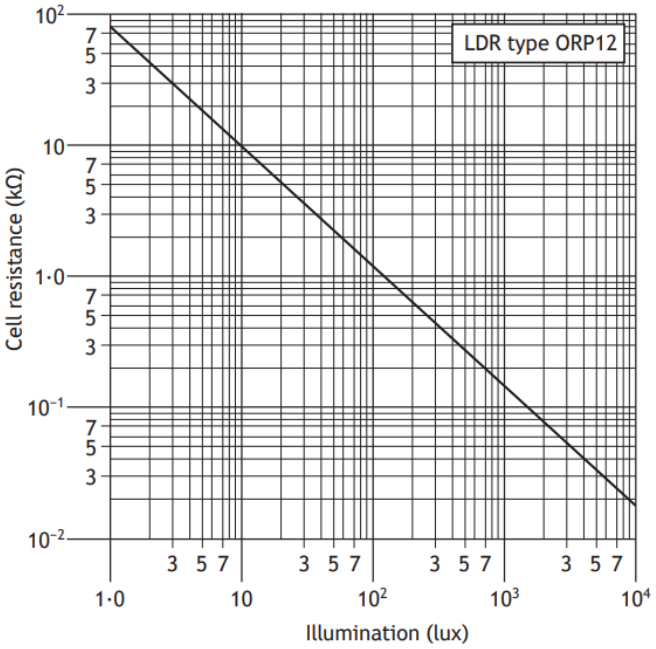
It can be seen from the diagram that

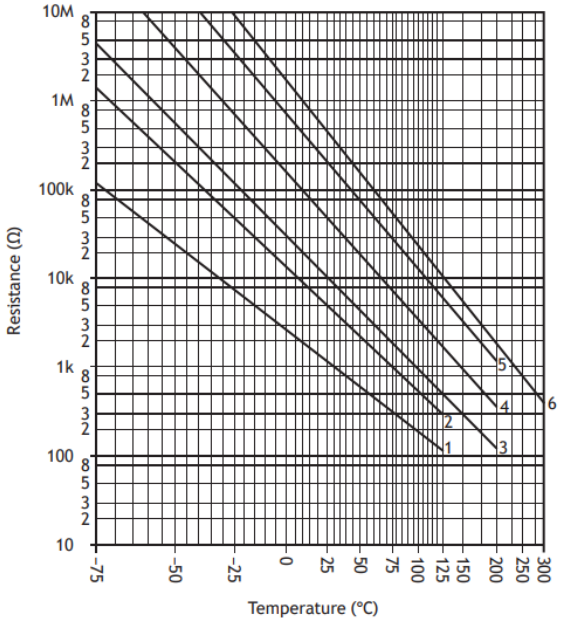
$V_b = V_{be} + V_e$
$V_{CC} = V_L + V_{ce} + V_e$



Transducer graphs

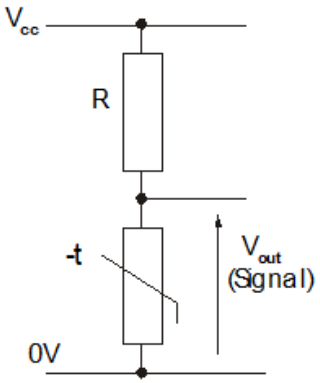
To work out the resistance of components like LDRs or thermistors, graphs of their resistance are used. In the Higher course, you may know only the lux (light level) or temperature of a circuit, so you must be able to read these and determine the resistance of these components. A copy of the graph will always be given in the question. Below are examples of these and at the back of this booklet you can find both in larger format for assistance with answering questions. Only ever use the supplied graph. **Large graph versions are at the end of this booklet**





Voltage dividers

A typical voltage divider is shown:



Voltage divider circuits work on the basic electrical principle that if two resistors are connected in series across a supply, the voltage load across each of the resistors will be proportional to the value of the resistors. In general, to calculate the voltage over any resistor in a series circuit, we can use the equation:

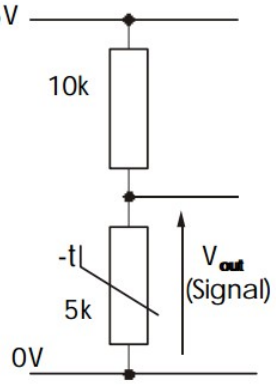
Where R_2 is the resistance of the thermistor in the above circuit and $R_T = R_1 + R_2$

When monitoring physical conditions, one of the resistors in the circuit is an input transducer, the resistance of which will change depending on the physical conditions.

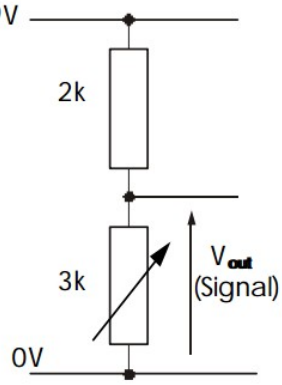
1

Copy the diagrams and calculate the signal voltages produced by the following voltage divider circuits:

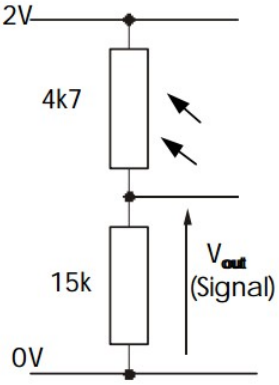
a)



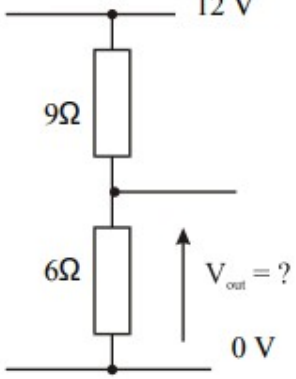
b)



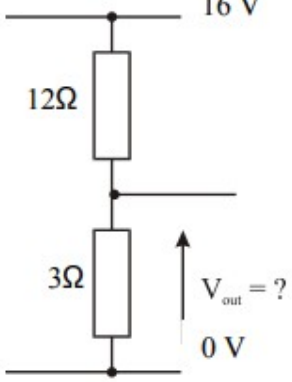
c)



d)



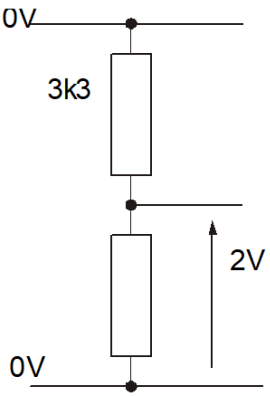
e)



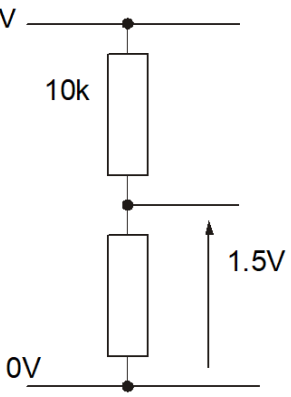
2

Copy the diagrams and calculate the size of the unknown resistor required for the given output voltages:

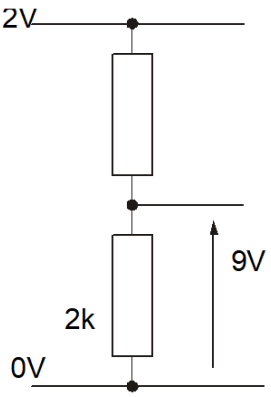
a)



b)



c)



Operational amplifiers

Operation Amplifiers (Op Amps) are the most commonly used devices that process the analogue signals. So far, the circuits and control systems we have looked at involve processing analogue signals such as heat, light, pressure or movement. Op Amps are used in these closed loop control systems.

They were originally designed to perform mathematical operations and used in analogue computers. They can be used to add, subtract, multiply, divide, integrate and differentiate electrical voltages. They can also amplify AC and DC signals.

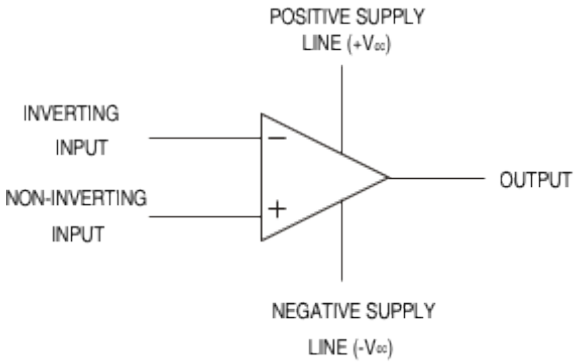
Op Amps are among the most widely used electronic devices today, being used in a vast array of consumer, industrial, and scientific devices. Many standard IC Op Amps cost only a few pence in moderate production volume; however, some integrated or hybrid operational amplifiers with special performance specifications may cost over £100 in small specialist quantities. Op Amps may be packaged as components or used as elements of more complex integrated circuits.

An "ideal" amplifier should have the following qualities:

- an infinite input resistance (typically 1M or more), so that very little current is drawn from the source (as V=IR);
- zero output resistance (typically 100 ohms or less), so that variations in load have very little effect on the amplifier output;
- an extremely high gain (typically 100,000);
- no output when the input is zero (in practice this is seldom achieved, however manufacturers provide an "offset - null" to compensate for this).

Although a typical Op Amp can contain more than 20 transistors and other components, we can treat it as a "black box" since we are only concerned with the input and output signals.

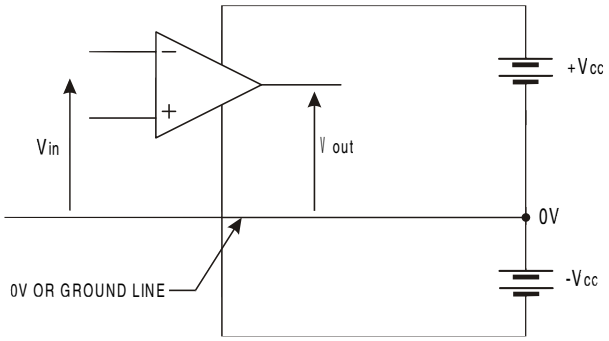
The symbol for an Op Amp is shown below:



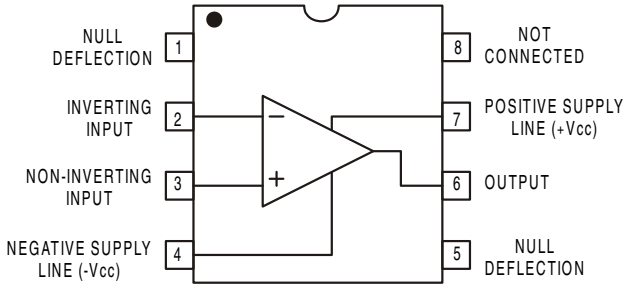
The Op Amp has two inputs and one output. The two inputs are indicated by a "-" and "+". A positive signal to the "-" input is amplified and appears as a negative signal at the output. Because of this, the "-" input is known as the inverting input (a negative signal would appear as a positive at the output). A positive signal to the "+" input is amplified and appears as a positive signal at the output. The "+" input is known as the non-inverting input. In addition, you can see Op Amps have their own external power supply.

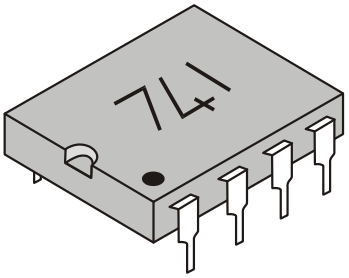
The output of an Op Amp can increase, but at a certain point the Op Amp is saturated and cannot increase further. It must be noted that the output of an Op Amp **cannot** be greater than 85% of its supply voltage. E.g. If an Op Amp has a +10V and -10V supply, it's output must lie somewhere between +8.5V and -8.5V.

Since the input and output signals can be either positive or negative, the Op Amp is usually powered from a dual rail supply and voltages measured relative to a zero volt (or ground) line as shown below. However it is normal practice to omit power lines when drawing diagrams as they are taken for granted.



Op Amp ICs come in two forms, the most popular of which is the DIL (dual in-line) package. The pin diagram for a 741 chip is shown below.





As you know, the top of any IC is usually indicated by a notch. Occasionally pin number 1 is indicated by a dot. Pins are always numbered from pin 1 in an anticlockwise direction. Connections to the offset null are usually made by means of a potentiometer. This will depend on the type of Op Amp used and reference should be made to appropriate data sheets if this is required.

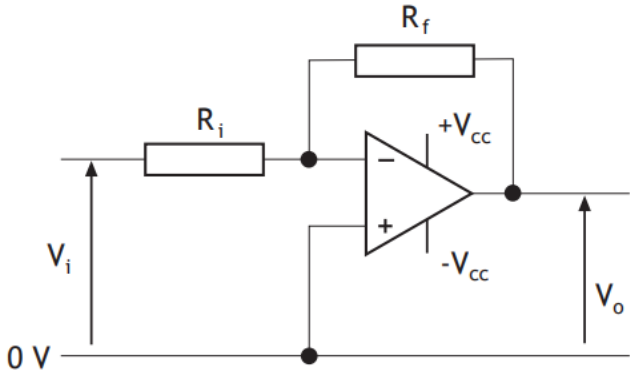
Op Amp Configurations

There are 6 Op Amp configurations you need to be aware of:

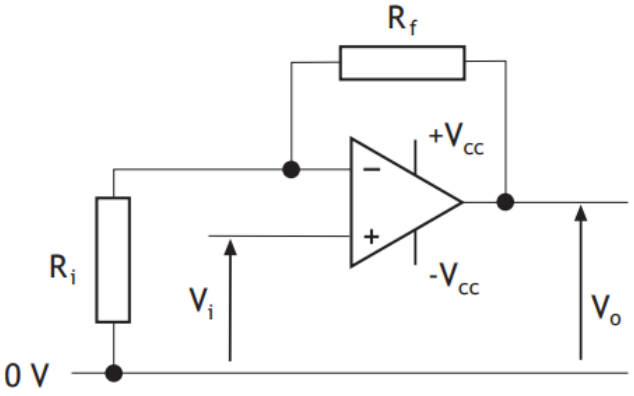
- Inverting amplifier
- Non-inverting amplifier
- Comparator
- Difference amplifier
- Summing amplifier
- Voltage follower

An example of each is shown in your Higher data booklet, along with the related formula. However, do not rely on this. It is vital you become familiar with each type and can identify it and the way it works. On the next pages you will see typical examples of each configuration with a brief explanation. You may use this as a reference sheet throughout this work, but get used to using your data booklet for answering questions.

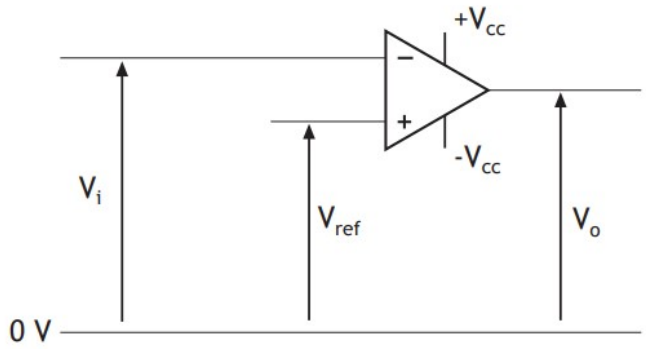
INVERTING AMPLIFIER – where the output is larger or smaller than the input and inverted (made negative)



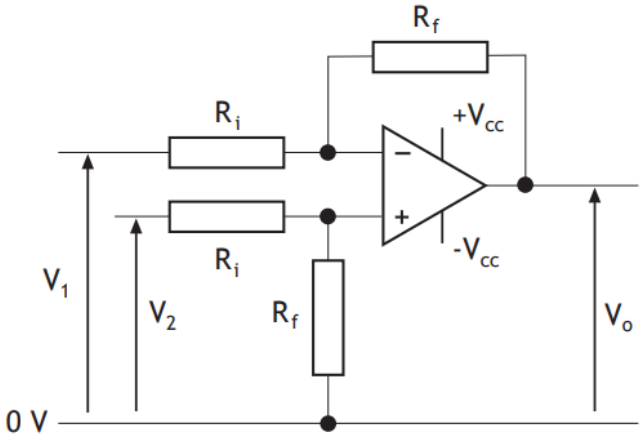
NON-INVERTING AMPLIFIER – where the output is larger than (or the same size as) the input.



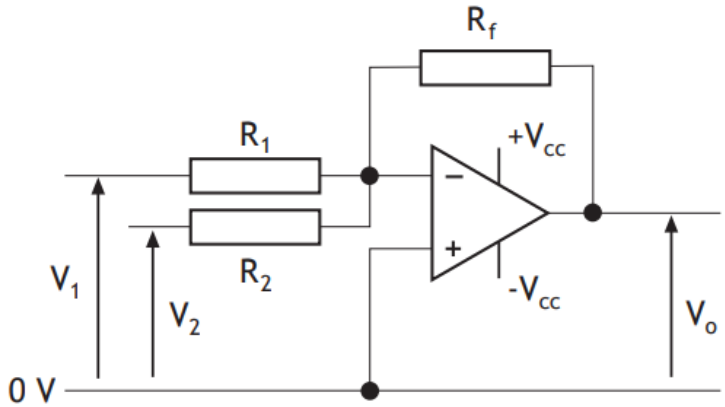
COMPARATOR – where the output switches on when an input is bigger than a reference voltage.



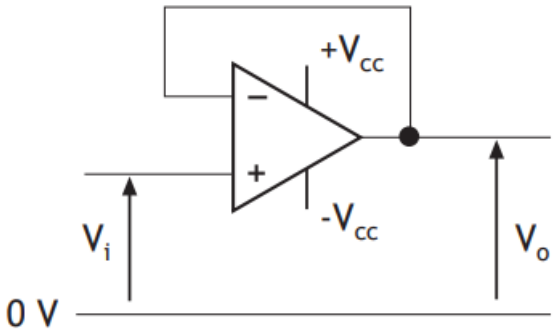
DIFFERENCE AMPLIFIER – where the output is an amplified version of the difference between the two inputs.



SUMMING AMPLIFIER – where a number of inputs are amplified and then added together to form a single output.



VOLTAGE FOLLOWER – where the output is exactly the same voltage as the input.



What is gain?

The Op Amp was designed as a voltage amplifier and when we are working with them, we are interested in how much bigger the output is compared to the input. This characteristic is called voltage 'gain', and is defined as:

$$Voltage \quad gain = \frac{Voltage \quad output}{Voltage \quad input}$$

$$A_V = \frac{V_o}{V_i}$$

Note that 'gain' is a 'ratio' and therefore has **NO** units. **'Gain' the is multiplier or ratio between the signal at the input V's the signal at the output.**

An Op Amp without and resistors connected (in comparator config.) has a gain of around 200 000, so even the smallest difference between the two inputs will result in a very large output. If there is a 1V difference, there can't be a supply of 200 000V output. As stated before, the maximum if determined by the size of the supply voltages (usually between 5 and 15V) and is a maximum of 85% of the voltage supplied.

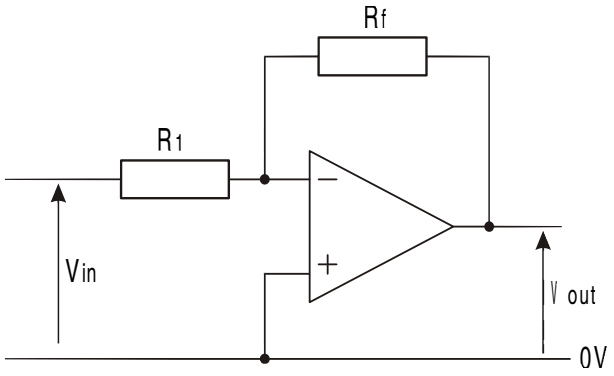
e.g. If a supply voltage is ± 15V the biggest output possible is ± 12.75V

The specific equation for gain changes depending on the configuration of your Op Amp, so be sure to check you are using the correct formula. You can always use your data booklet for this.

The inverting amplifier configuration

If we want a signal to be inverted, we use the inverting configuration. A positive input signal would be converted to an amplified negative signal, and vice versa. This is useful for negative feedback scenarios, such as a heating system or a lighting system where lights dim or brighten with daylight conditions.

The input signal, normally from an input transducer, is connected to the inverting input through an input resistor (R_1). Negative feedback is provided by linking the output signal to a feedback resistors R_f. The non - inverting input is connected to ground (0V) either directly or through a biasing resistor R_b. If used, R_b should have the equivalent resistance as R_1 and R_f connected in parallel.

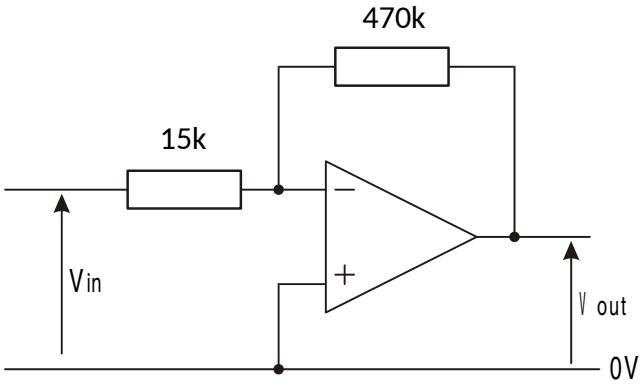


$$A_v = \frac{V_o}{V_i} \qquad A_v = -\frac{R_f}{R_i} \qquad V_o = -\frac{R_f}{R_i} V_i$$

Gain A_v will always be negative. Note that the input resistance of the circuit = R_1.
If V_i is a positive voltage then V_o will be negative. V_o can be bigger, smaller or the same size as V_i.

Worked example

An Op Amp is used in a circuit as shown to the right, with R_1 = 15k and R_f = 470k. Calculate the gain of the circuit and determine the output voltage when an input signal of 0.2V is applied.



Step 1

Calculate the gain (as it is a ratio, NO UNITS)

$$A_v = -\frac{R_f}{R_1} = -\frac{470k}{15k} = \underline{-31.33}$$

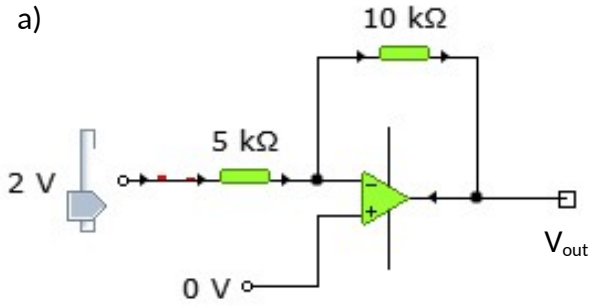
Step 2

Calculate the output voltage

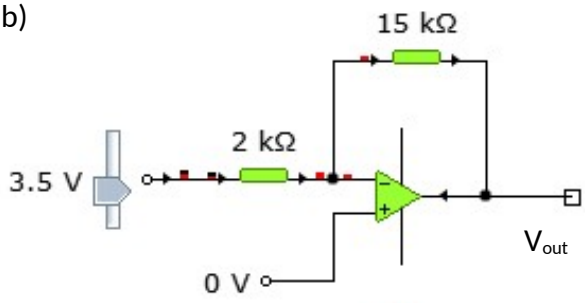
$V_o = A_v \times V_i = -31.33 \times 0.2 = \underline{-6.266\ V}$

3 For the following inverting op-amps, calculate the missing quantity.

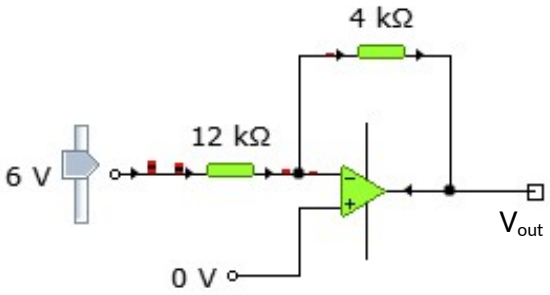
a)



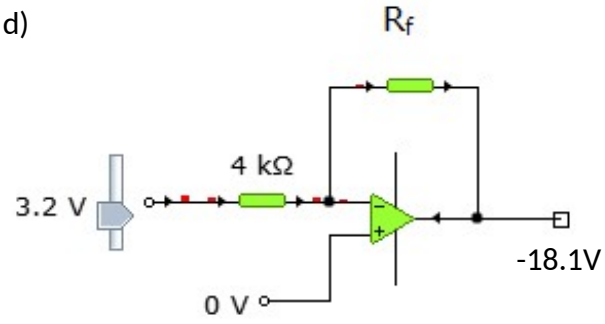
b)



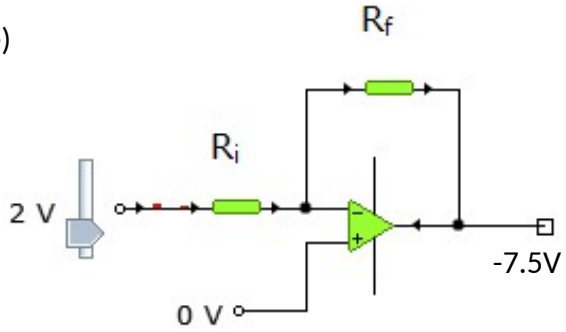
c)



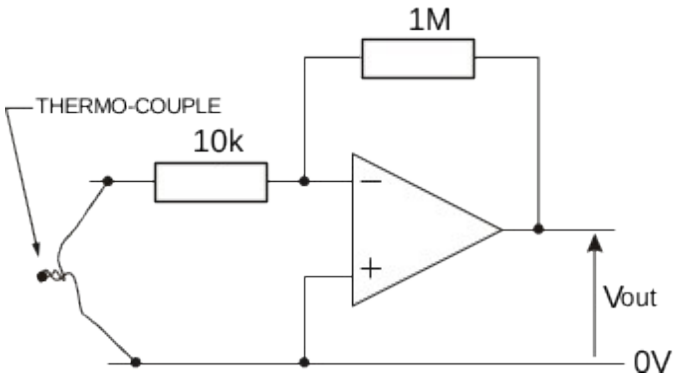
d)



e)



4 A thermocouple known to produce an output of 40 micro volts per °C is connected to an Op Amp circuit as shown below:



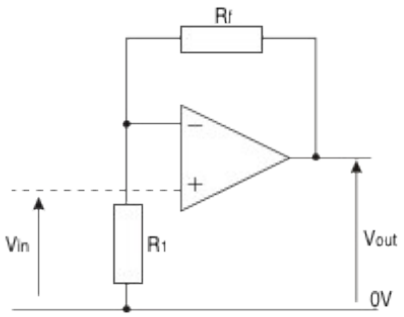
a) Calculate the gain of the circuit.
b) Determine the output voltage if the thermocouple is heated to a temperature of 1000 °C.

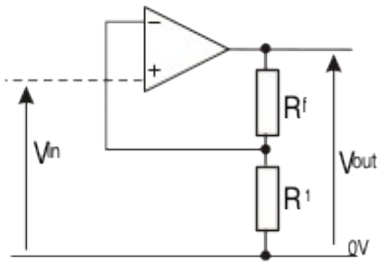
Note: For an inverting amplifier, the sign of the output voltage is the opposite of the input voltage. In order to obtain the same sign, the output signal could then be fed through another inverter (with $R_f = R_1$, so that the gain = -1).

The non-inverting amplifier configuration

In this configuration, the input voltage signal V_i is applied directly to the non-inverting (+) terminal. The means the gain will be positive and the input signal will be multiplied but not inverted. A non-inverting configuration is used when a weak signal needs to be made bigger. An example of this would be as a first stage amplified after the microphone in your mobile phone.

R_f and R_1 form a voltage divider circuit feeding back some of the output signal to the inverting input. Below are two diagrams showing two different ways of drawing the same circuit. Note that the input resistance of the circuit is very high as it is the input resistance of the Op Amp, making it very useful when input transducers do not provide large enough currents to drive desired outputs.



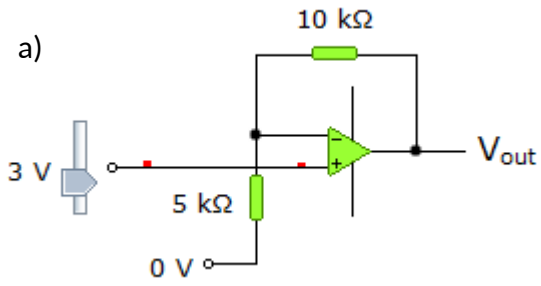


$$A_v = \frac{V_o}{V_i} \qquad A_v = 1 + \frac{R_f}{R_i} \qquad V_o = \left(1 + \frac{R_f}{R_i}\right) V_i$$

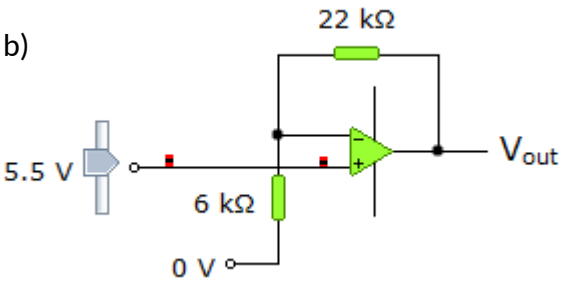
5

Calculate the missing quantity for the following non-inverting op-amps.

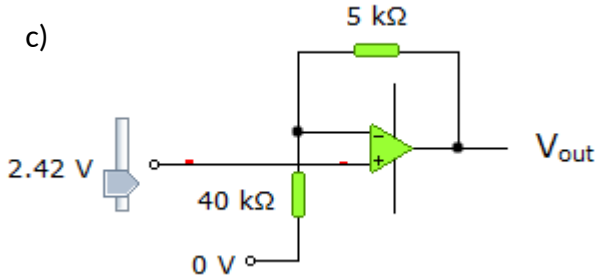
a)



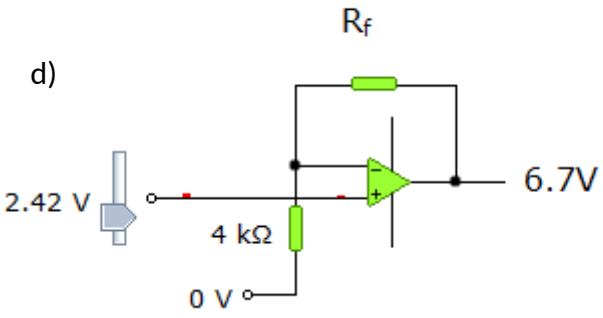
b)



c)

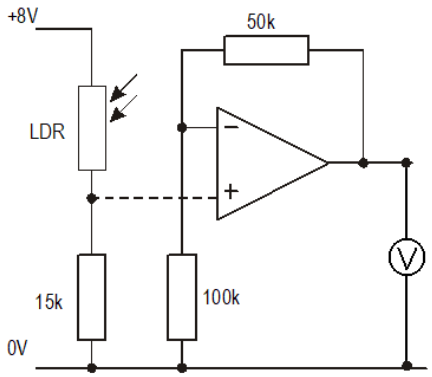


d)



6

To build a simple light meter, a light dependent resistor (LDR) is connected into a circuit as shown below:

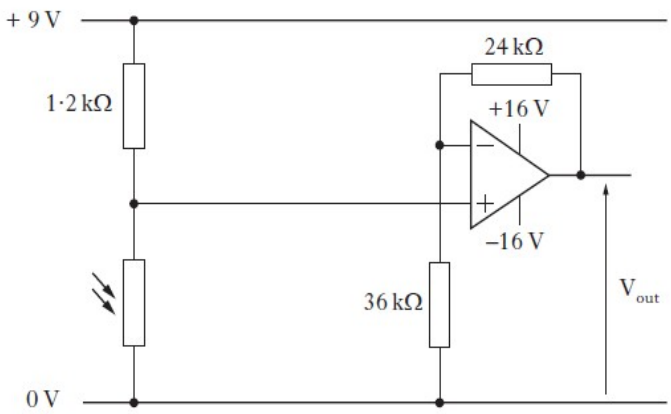


In bright sunlight, the LDR has a resistance of 1k. In shade, it's resistance increases to 15k.

a) Determine the voltages that would appear on the voltmeter in both light conditions.
b) How could the circuit be altered to indicate changes in temperature?

7

The circuit shown below is used in a light-sensing system for a camera:

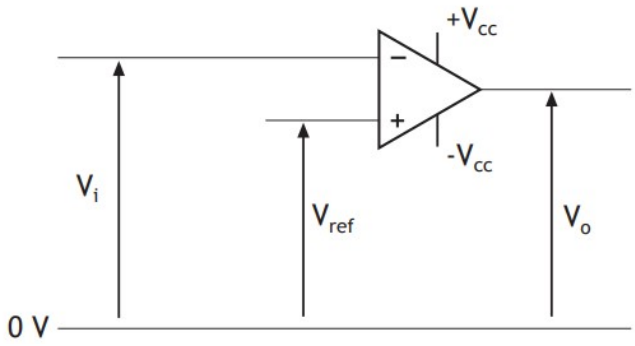


a) Calculate the output voltage V_o when the light level is 80 lux. Use the LDR graph on page 4.
b) Determine the minimum light level that this system can respond to before the Op Amp saturates.

The comparator configuration

In its most simple configuration, an Op Amp will act as a comparator. This means it will compare two separate voltage signals, V_i and V_{ref}, where V_i is the input signal coming from an input transducer and V_{ref} is the reference voltage used for comparison. This is commonly used in control circuits in which loads are switched on and off and is an open loop, with no feedback. The gain of the circuit is therefore A_0 and any small difference in the two input signals is amplified so the Op Amp saturates (either positively or negatively).

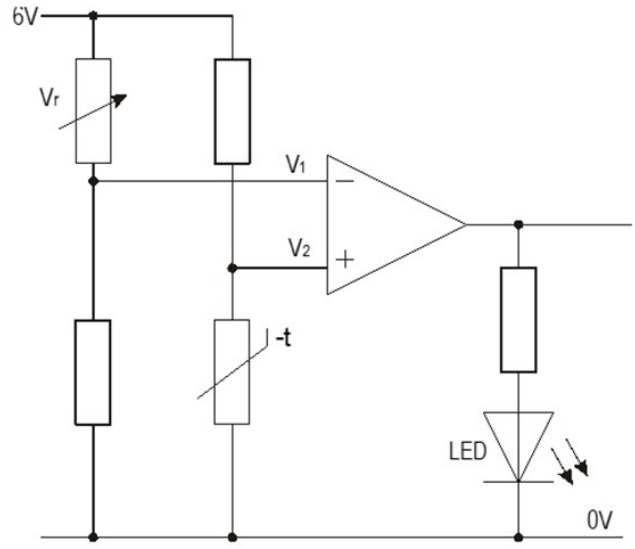
Within a comparator, if V_i is higher than V_{ref} the output V_o is 'low' (this means that it will have a negative output). If V_i is lower than V_{ref} the output will be 'high' (meaning there will be a positive output). If both inputs are exactly the same, then the output should be zero.



$A_v = A_0$
If $V_i < V_{ref}$, then Vo saturates positively (85% of $+V_{cc}$)
If $V_i > V_{ref}$, then Vo saturates negatively (85% of $-V_{cc}$

Example

The circuit shown below would give an indication when the temperature falls below a pre-set value (0°C for example).



V_r is adjusted until V_1 is just greater than V_2, the output will therefore be negative and the LED will be off. This sets the reference value for the system.

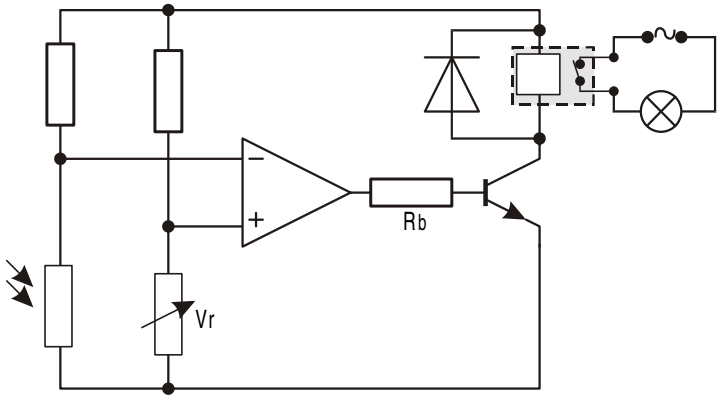
As the temperature falls, the resistance of the NTC thermistor rises and therefore V_2 gets bigger. Eventually when it is larger than the voltage at the inverting input, $V_2 > V_1$, the Op Amp will saturate, the output goes positive and the LED lights.

In simple terms;
If the non-inverting input has a higher value the output saturates positive.
If the inverting input has a higher value then the output saturates negative.

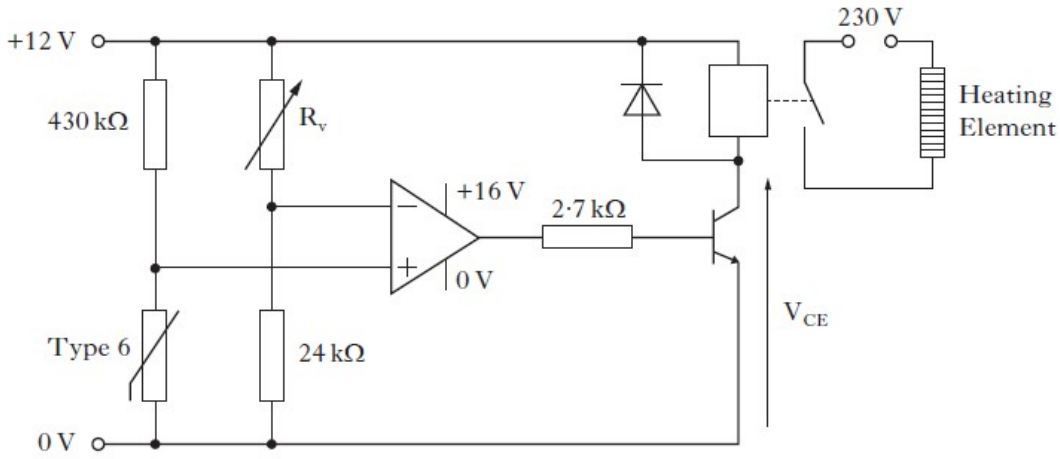
Driving external loads

The maximum output current that can be drawn from an Op Amp is usually low (typically 5 mA). If larger currents are required, the output could be connected to a transducer driver either a bipolar transistor or MOSFET (and relay circuit if required). You will learn more about transistors in the next part of this course.

8 Describe the operation of the circuit shown below and state the purpose of the variable resistor V_r and the fixed resistor R_b. (For clarity, the DC power supply has not been shown). (6 marks)

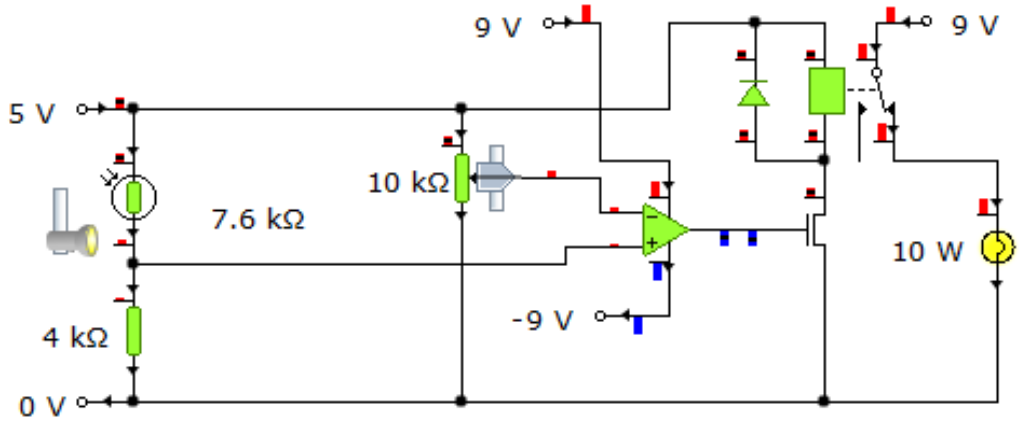


9 A room heating system uses the control circuit shown below. A thermistor monitors the room temperature.



a) State the full name of the type of control used.
b) State the function of the diode in this circuit.
c) Explain clearly the operation of the circuit as the temperature falls. The heating element should switch on at 16 °C. (4 marks)

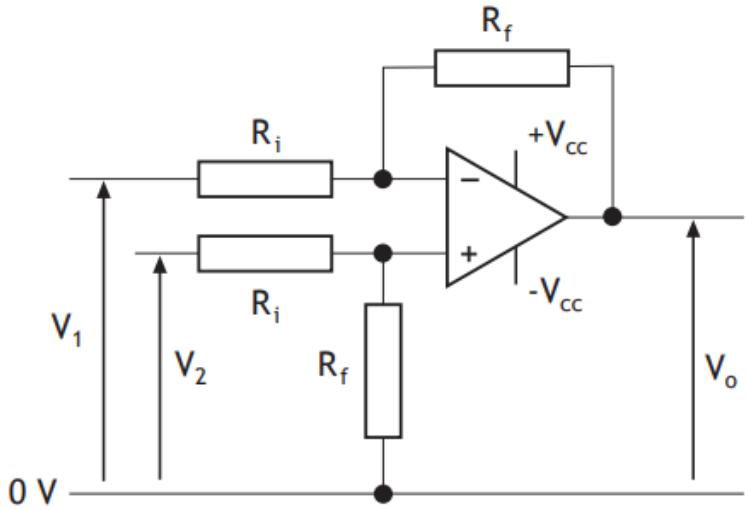
10 Describe the operation of the circuit shown below and state the purpose of the potentiometer.



The difference amplifier configuration

In this configuration the Op Amp amplifies the difference between the two signals. If there is no difference between the input signals the output with be 0. When a difference exists the output switches on and the amount the signal is amplified by depends on the difference found.

This configuration is used to produce proportional control; if a large difference exists the output will be large, if the difference is small the output signal will be small.

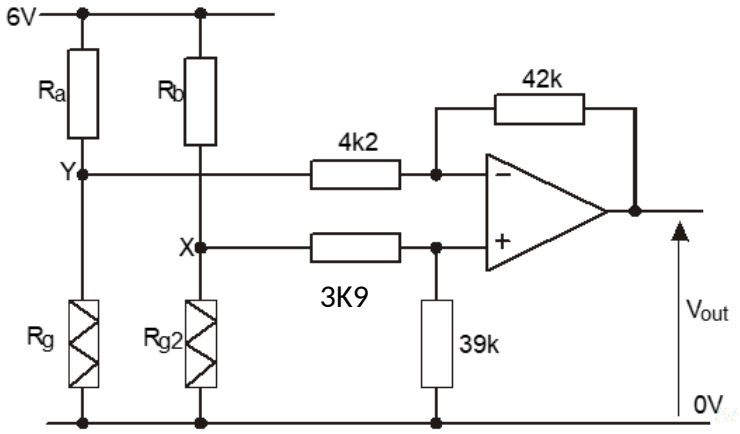


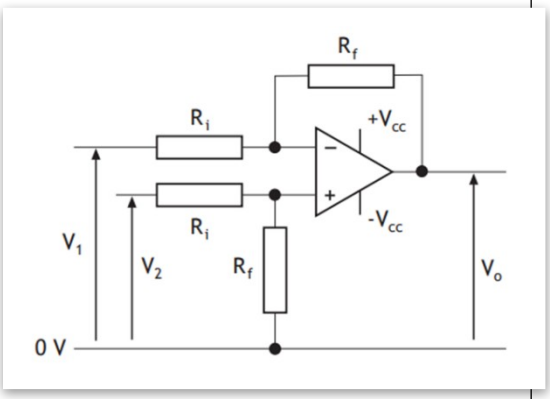
$$A_v = \frac{V_o}{(V_2 - V_1)} \qquad A_v = \frac{R_f}{R_i} \qquad V_o = \frac{R_f}{R_i}(V_2 - V_1)$$

Note: To ensure each input is amplified by the same amount, the circuit is designed with the same resistor values, so their ration is identical.

11

Two strain gauges are connected to a **difference amplifier** as shown below. Note: Strain gauges are sensors which convert force into electrical resistance. Use you data booklet for the equations.





$R_A = R_B = 1\ k\Omega$, when not under strain, $R_{g1} = R_{g2} = 200\Omega$

a) Calculate the voltage at X and Y when both gauges are not under strain and hence determine the output voltage of the amplifier.

b) As the strain of R_{g2} increases, its resistance increases from 200 to 210 Ω, determine the new output voltage.

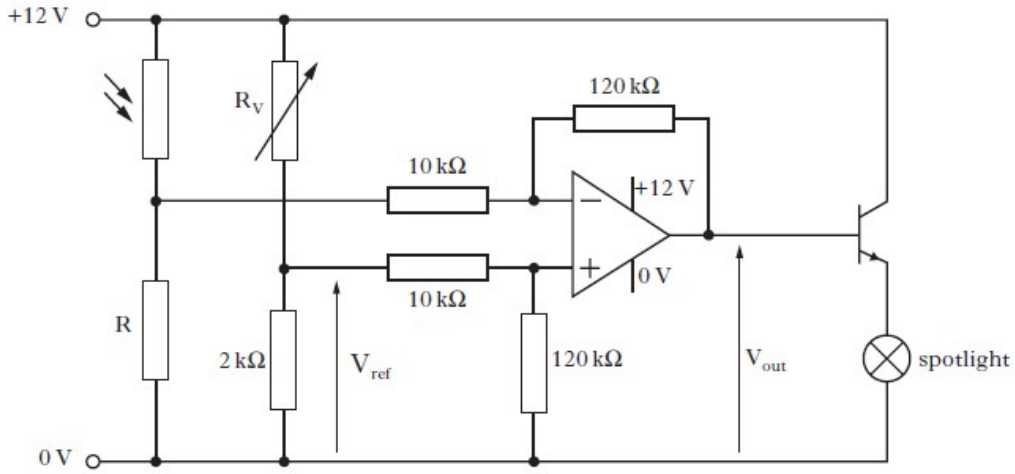
c) What would you expect to happen to the output voltage if both gauges were put under the same amount of strain?

12

An electronic system controls a spotlight in an art gallery. As the natural light level varies, the spotlight brightens or dims as required, in order to keep the overall light level constant. The light level at which the spotlight is triggered can be adjusted.

a) Draw a control diagram for the system.

The op-amp-based circuit for the system is shown below:

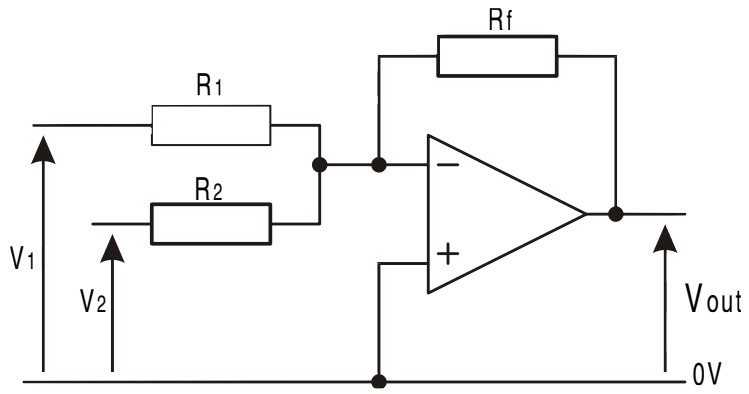


b) Calculate the voltage V_{ref} when the variable resistor $R_v = 4{\cdot}8k\Omega$

c) Calculate the resistance R when the light level is 400 lux and V_{out} = 8V. Note: The Vout position for this question is <u>before</u> the op amp and not where it is illustrated on the diagram after the op amp.

The summing amplifier configuration

In this configuration the Op Amp combines the voltages of two or more inputs into one single output. This configuration is used in a wide range of electronic circuits, for example audio mixers in recording studios. The summing amplifier adds the waveforms from the various channels (guitars, vocals, drums, etc.) before sending the mixed signal to a recording device.



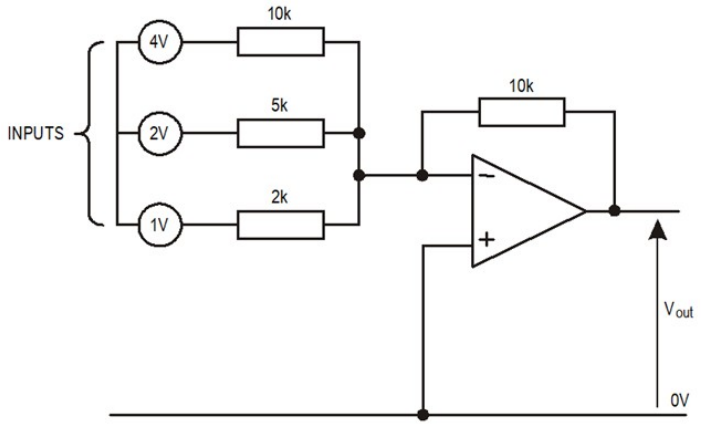
$$A_{v1} = -\frac{R_f}{R_1} \qquad A_{v2} = -\frac{R_f}{R_2} \qquad A_{vn} = -\frac{R_f}{R_n}$$

$$V_o = (A_{v1} V_1) + (A_{v2} V_2) + \ldots$$

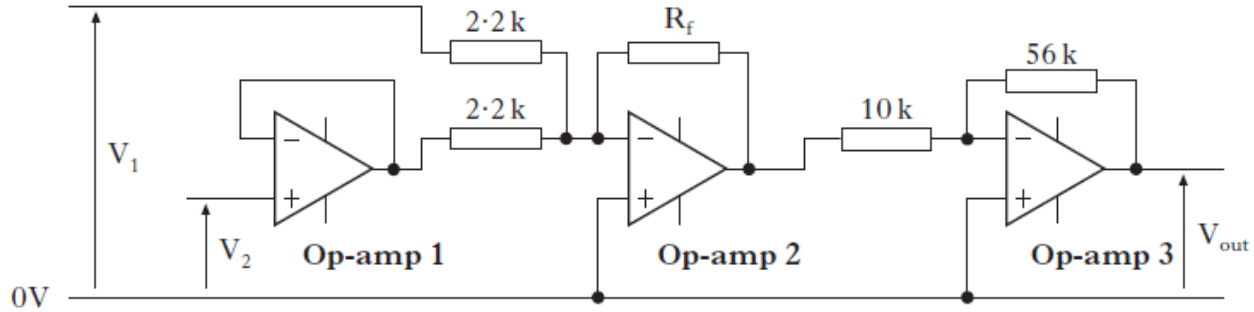
$$V_o = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \ldots \right)$$

Any number of inputs can be added in this way and Rf affects the gain of every input.

13 Determine the output voltage for the circuit shown below.



A signal conditioning circuit for a load cell output voltage V_1 is shown below an adjustable input voltage V_2 enables the output voltage V_o to be set to zero when there is no load on the load cell.



Explain the purpose of Op Amp 1 in this circuit. The table below shows one set of readings taken when calibrating the load.

LOAD (kN)	V_1 (mV)	V_2 (mV)	V_{out} (V)
22·0	120	–5	4·391

b)

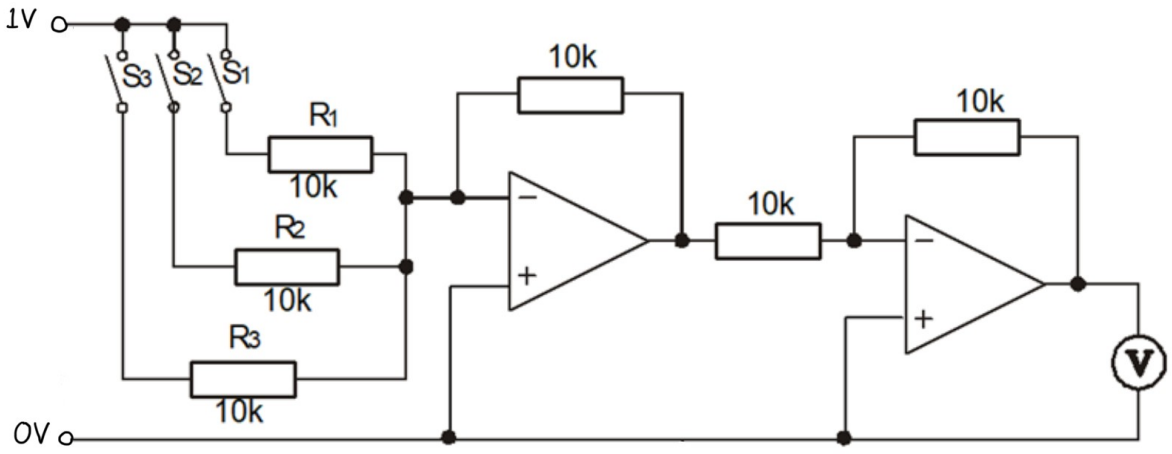
Calculate the value of the feedback resistor R_f used in the signal conditioning circuit.

c) Explain what it means to *calibrate* the load sensor.

Circuit Simulation with Yenka

15 Digital-to-analogue converter

Digital devices produce ON/OFF signals. Processing takes place using the binary number system (as oppose to the decimal system). Construct the circuit shown below in Yenka.



Since all inputs are amplified by the same amount (same value of input resistors), therefore:

The output voltage = $\sum$ Input voltages.

Example. if S1, ON (connected to 1V) and S2, ON (connected to 1V), the output voltage should = (1 + 1) = 2V

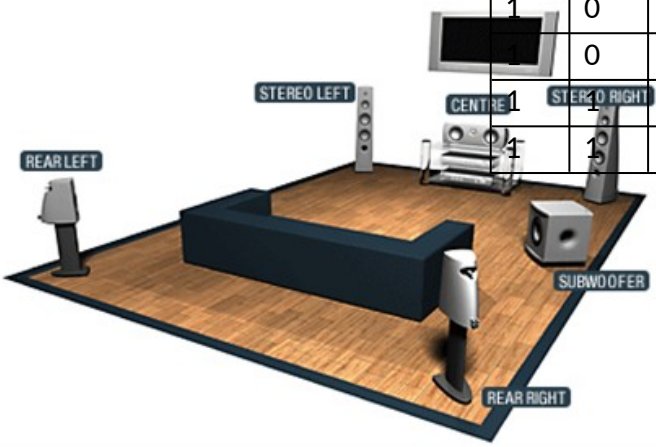
Now change the circuit so that R2 = 5k and R3 = 2.5k. Complete the table to show the state of the input switches and the output voltage.

16 Surround sound system

The amplifier for a surround sound system receives signals from a set-top box and a DVD player which produce output signals of 50 mV and 10 mV respectively.

Design a circuit so that an output of 1 volt is produced when either of the inputs are used. Use Yenka to model your circuit and print and paste into your jotter.

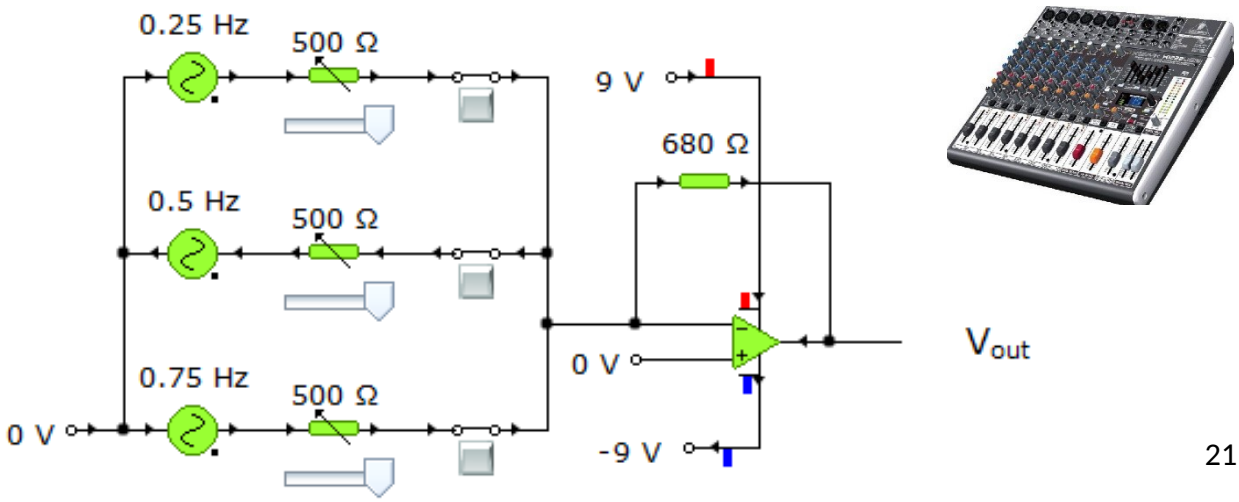
S1	S2	S3	output voltage (V)
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	



17 AC mixer pre-amplifier

Mixers allow different signals to be amplified by different amounts before being fed to the main amplifier. Signals might come from microphones, guitar pick-ups, pre-recorded sound tracks etc.

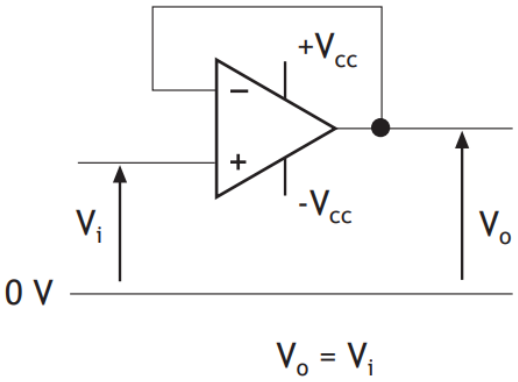
Construct the circuit shown below on Yenka. Use the graph function to monitor the output voltage. Adjust the frequencies of the signals as shown and adjust the signal generators to give a maximum voltage of 1V and a minimum of -1V.



- Putting each switch on individually will allow you to “see” each of the input signals in turn.
- Putting more than one switch on at a time will show you the sum of the input signals.
- Adjusting the size of the input variable resistors alters the amplification of that particular input signal.
- (Complex output signals can be constructed by adding sine waves of the correct amplitude and frequency - useful in electronic keyboards or synthesisers when a particular musical instrument is required.)

The voltage follower configuration

This is a special case of the non-inverting amplifier in which 100% negative feedback is obtained by connecting the output directly to the inverting input.



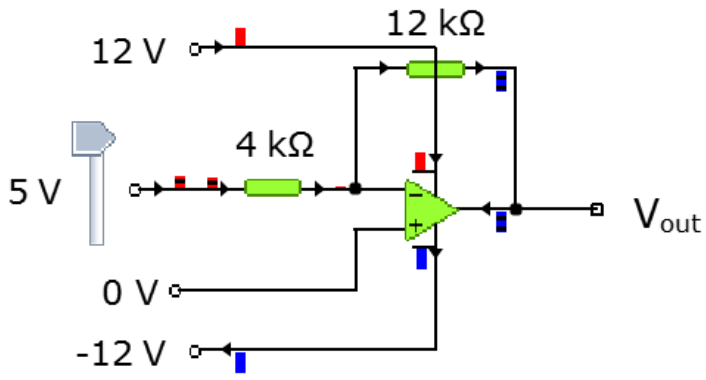
$V_o = V_i$

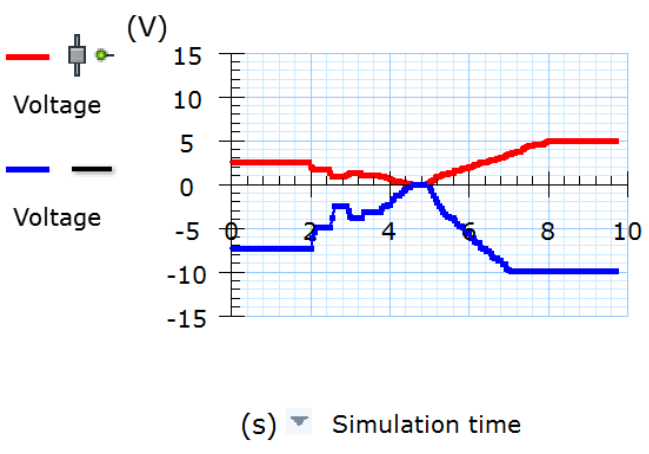
Since $R_f = 0$, the gain of this circuit is 1 i.e: The output voltage = Input voltage.

This configuration is commonly used to protect sensitive input sensors or prevent their operation being affected when connected to transistors or other processing parts of a control system. This is sometimes referred to as an input buffer, where an input signal has a high voltage but low current. An example of this could be a signal from a microphone sent to drive a loud speaker.

Circuit simulation with Yenka

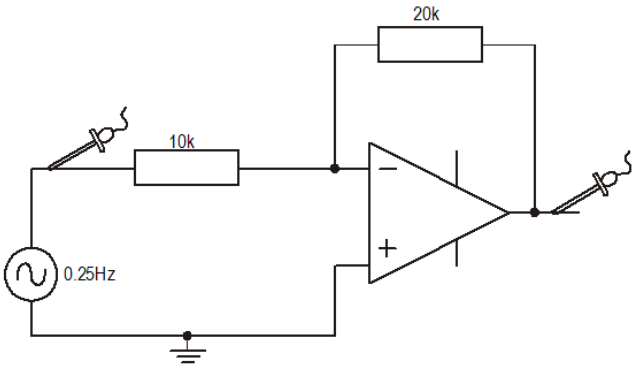
18 Construct the circuit shown below. Use the graph to monitor what happens to V_{out} as the input voltage is varied. Explain what happens as the input reaches its maximum.





19

Construct the circuit shown below.



Voltage (Volts)

10									
0	2	4	6	8	10	12	14	16	18
-10									

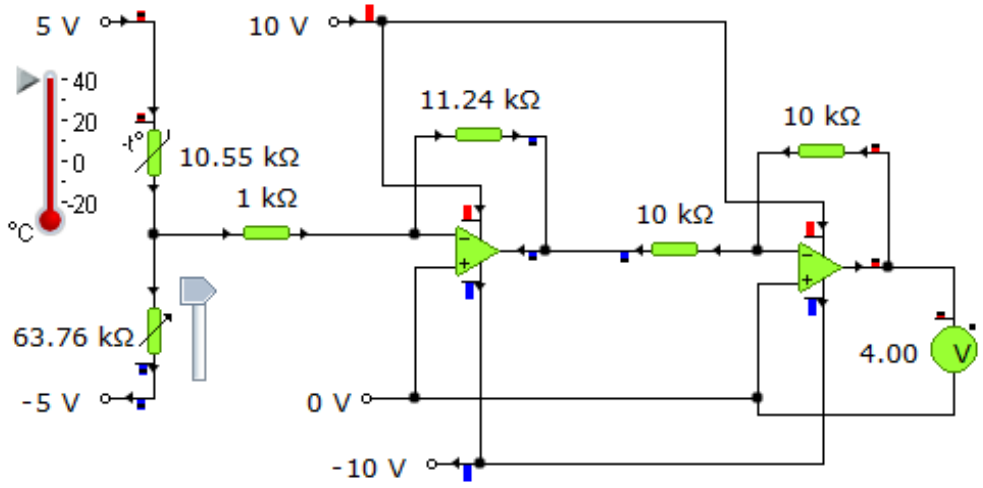
Time (seconds)

Set the input voltage to 2 Volts, 0.25 Hz. Set the oscilloscope to a maximum voltage of 10 V and a minimum voltage of -10 V. Start the trace on the oscilloscope and compare the input and output voltages.

Now increase the size of the feedback resistor to 50k and repeat the exercise. This time you should observe "clipping" of the output signal.

2

Construct the "electronic thermometer" circuit shown below. This uses an inverting amplifier and a voltage follower.



Set the "temperature" to 0°C, adjust the variable resistor (V_r) until the voltmeter reads 0.00V. Increase the "temperature" to 40°C, adjust the feedback resistor in the inverting amplifier until the voltmeter reads 0.40V. The electronic thermometer is now calibrated to read 0.00 at 0°C and 0.40 at 40°C.

Investigate voltage readings at various other "temperatures" and suggest why this would not make a good thermometer.

21 Choose a suitable op-amp configuration for each of the following situations.

Situation	Configuration
An output voltage must be larger than the input.	
When temperature rises above a set level an output device will switch on.	
An analogue input signal is fluctuating, and a steady output voltage is required.	
The voltages from three input sensors must be combined to produce a single output voltage.	
Two input voltages must be compared. The output voltage must be proportional to the difference between them.	
An output signal must be a larger, but negative, version of the input.	
An output signal must be a smaller, but negative, version of the input.	
The output voltages from 4 pins on a microcontroller must be combined, in different ratios, to produce a single output signal.	

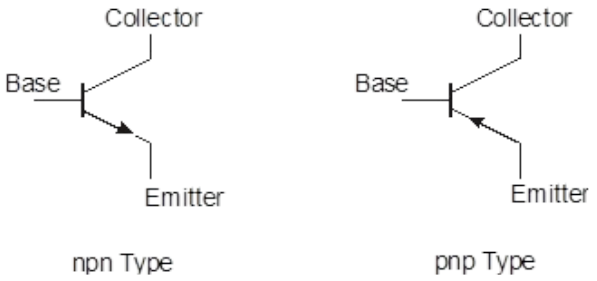
Transistors

Input transducers that produce voltage rarely produce sufficient voltage for most applications. This means their outputs must be amplified.

Amplifying devices are said to be active components, as oppose to non-amplifying components (resistors, capacitors etc.) which are known as passive components. The extra energy required to operate the active component comes from an external power source (battery, transformer, etc.).

BJTs

The most common active device in an electronic system is the bipolar junction transistor (BJT). Two types are available, NPN or PNP.



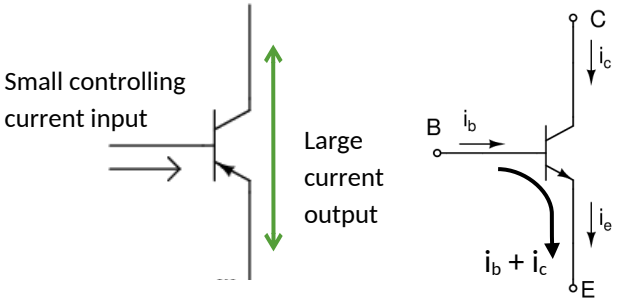
npn Type pnp Type

The transistor must be connected into circuits correctly. The arrow head on the emitter indicates the direction of conventional current flow (positive-to-negative). In this course only NPN transistors will be considered, but the difference between them is the NPN operated when the base is made positive and the PNP operates when the base is made negative.

A small current flowing between the base and emitter of a transistor allows a large current to flow between the collector and emitter, such that

$$I_e = I_b + I_c$$

Since I_b is usually much smaller than I_c, it follows that I_e is approximately = I_c. In the common emitter mode, a small current flowing between the base and emitter junction will allow a large current to flow between the collector and emitter.



The current gain (or amplification) of the transistor is defined as the ratio of collector to base current and is denotes h_{FE}.

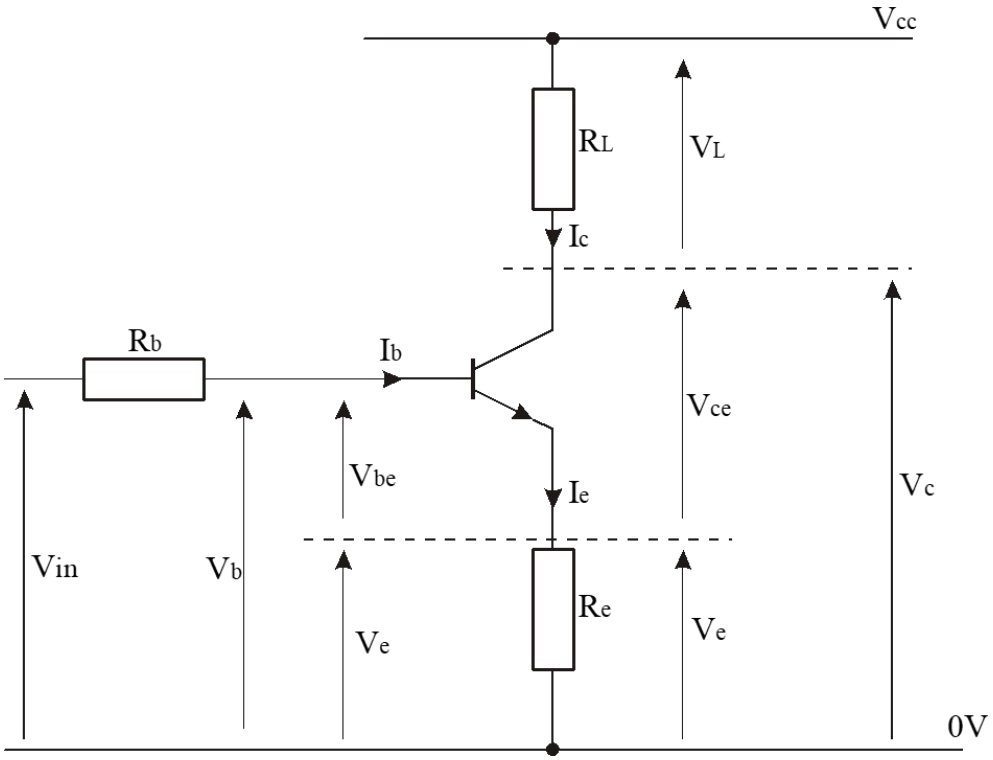
$$h_{fe} = \frac{\Delta I_c}{\Delta I_b}$$

In practice, the maximum allowable currents will depend on the make of transistor used. These limits can be obtained from manufacturers' data sheets. Forcing the transistor to carry currents greater than these maxima will cause the transistor to overheat and may damage it.

Transistor notation

The following may be useful as a reference page as you work through the questions.

I_c	- Collector current
I_b	- Base current
I_e	- Emitter current
V_{CC}	- Voltage of supply (relative to ground line)
V_b	- Voltage at the base junction (relative to ground line)
V_e	- Voltage at the emitter junction (relative to ground line)
V_{ce}	- Voltage between the collector and emitter junction
V_{be}	- Voltage between the base and emitter junction
V_L	- Voltage over the load resistor
Av	- Current gain
h_{FE}	- Hybrid parameter Forward current gain common Emitter



It can be seen from the diagram that

$V_b = V_{be} + V_e$

$V_{CC} = V_L + V_{ce} + V_e$

22

a) Calculate the gain of a transistor if the collector current is measured to be 10 mA when the base current is 0.25 mA.

b) Calculate the collector current through a transistor if the base current is 0.3 mA and hFE for the transistor is 250.

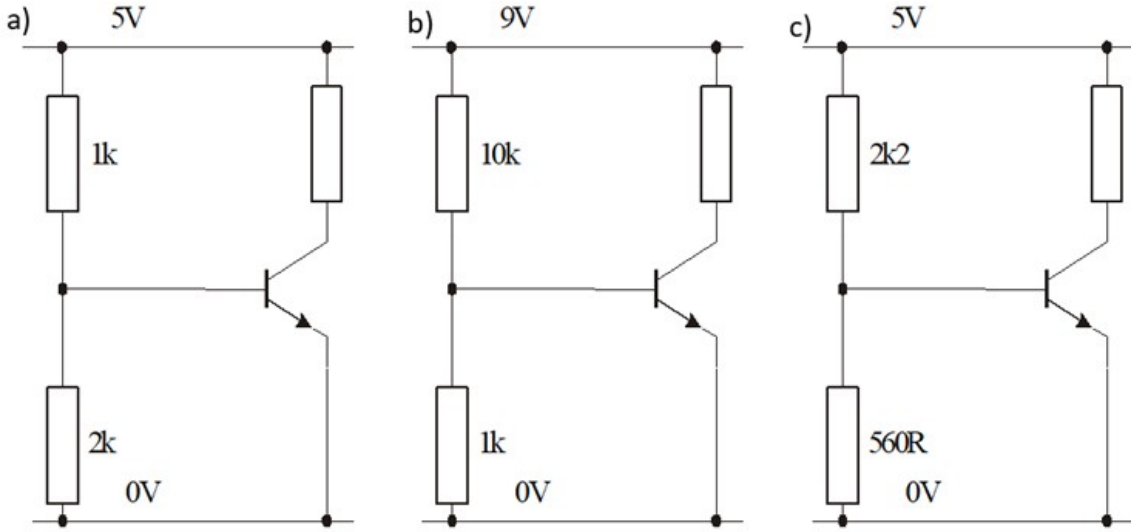
c) What collector current would be measured in a BC107 transistor if the base current is 0.2 mA and hFE is 100?

Transistor switching circuits

To generate a current in the base of the transistor, a voltage must be applied between the base - emitter junction V_{be}. It is found that no (or negligible) current flows in the base circuit unless V_{be} is above 0.6V. Increasing the base - emitter voltage further, increases the base current (producing a proportional increase in the collector current).

When the base - emitter voltage reaches about 0.7V, the resistance between the base emitter junction starts to change such that the base - emitter voltage remains at about 0.7V. At this point the transistor is said to be saturated. Increasing the base current further has no effect on the collector current and the transistor is fully on. It can be assumed that if the transistor is turned on, V_{be} = 0.7V

23 For each of the circuits shown, copy the diagram and calculate V_{in} and state if the transistor is ON or OFF (saturated).



Worked example

Consider the circuit shown below.

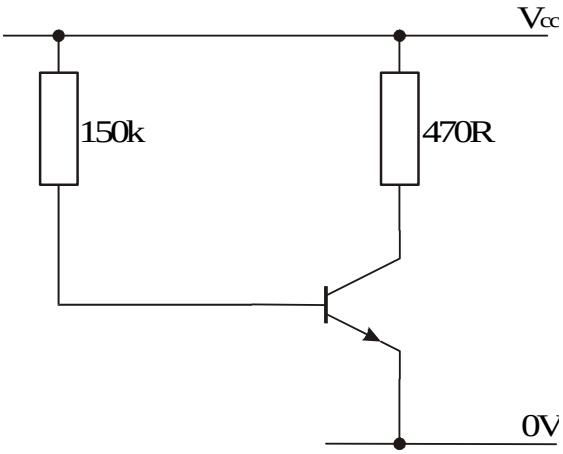
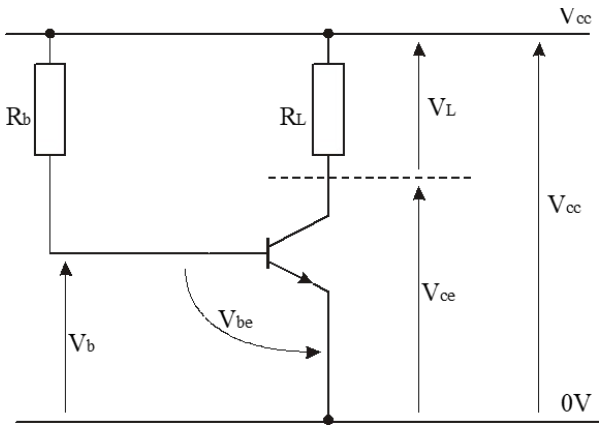


Diagram A



If the transistor is ON, calculate the collector current and V_{ce} , if h_{FE} =200 and V_{CC} = 9 Volts. Diagram A shows the circuit currents and voltages using standard notation.

Step 1

The voltage between the base and emitter junction is always about 0.7V. Since the emitter is connected to the ground line (0V), V_b= 0.7V.

Step 2

The voltage dropped over the base resistor can then be calculated

Voltage drop = V_{CC} - V_b = 9 - 0.7 = 8.3V

Step 3

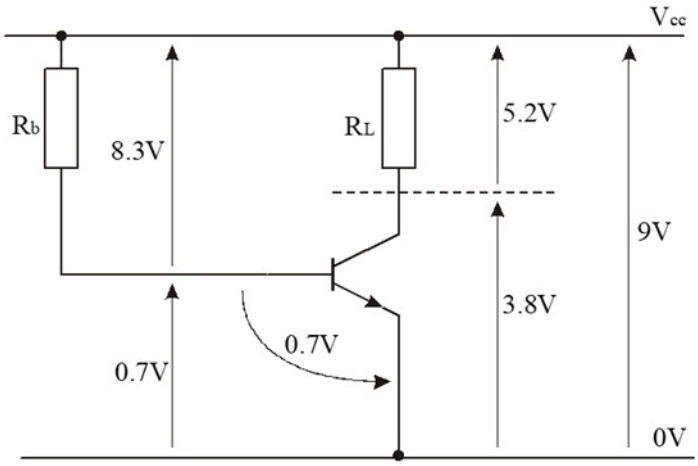
The base current is calculated using Ohm's law

$$I_b = \frac{V_{dropped}}{R_b} = \frac{8.3}{150k} = 0.0553\text{mA}$$

Step 4

I_c is calculated knowing h_{FE}

I_c = h_{FE} x I_b = 200 x 0.0553 = <u>11.06 mA</u>



Step 5

V_L is calculated using Ohm's law

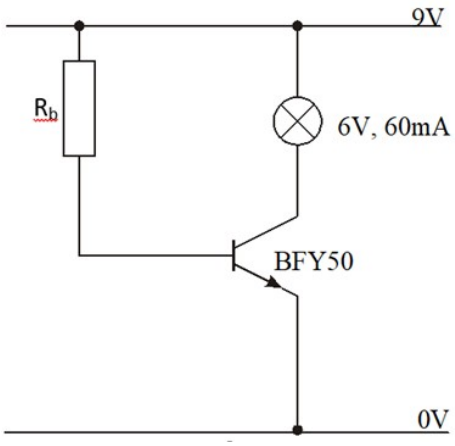
V_L = I_c x R_L = 11.06 mA x 470 = 5.2 V

Step 6

V_{ce} is calculated

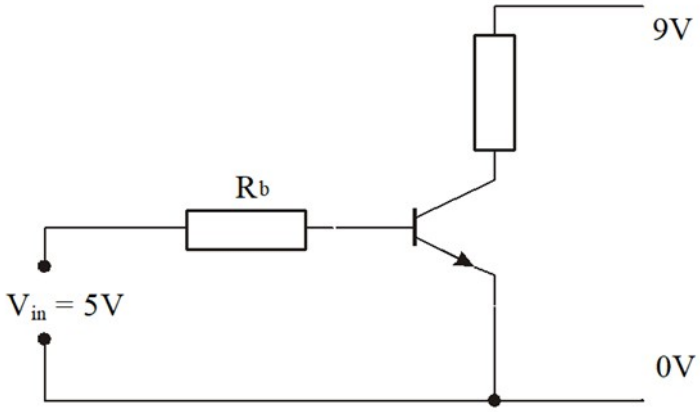
V_{ce} = V_{cc} - V_L = 9 - 5.17 = <u>3.8 V</u>

2 A 6 V, 60 mA bulb is connected to the collector of a BFY50 transistor as shown.



If the gain of the transistor is 30, determine the size of the base resistor R_b required to ensure that the bulb operates at its normal brightness.

25 Assume $I_{c(max)}$ for the transistor shown below R_b is 100 mA and h_{FE} is 200.



Calculate:

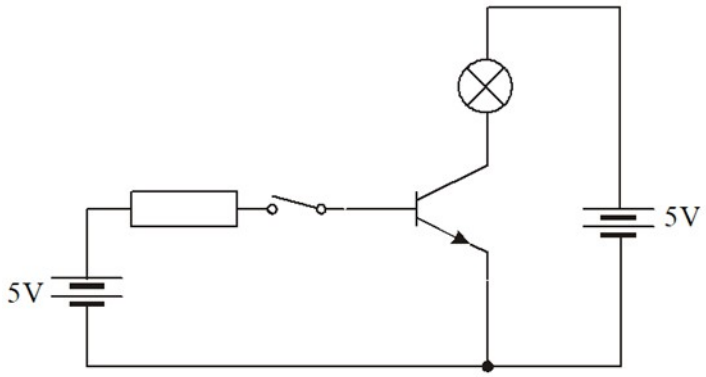
a) the maximum allowable base current.

b) the size of protection base resistor required (remembering V_{be} = 0.7V, and R = V/I)

Circuit simulation software

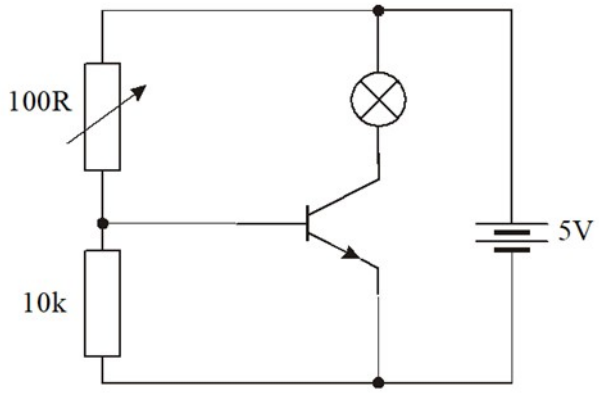
NOTE: THESE CIRCUITS INVESTIGATE THE NEED FOR PROTECTION IN ORDER TO AVOID DAMAGE. DO NOT CONSTRUCT THESE CIRCUITS USING "REAL" COMPONENTS.

26 Using the simulation software, construct the circuit shown in figure 15 using a 5 V supply. Switch on and see what happens.



Now insert a 10k base protection resistor and see what happens when you switch on now. Use the simulation to determine the smallest value of resistor required to protect this transistor.

27 Construct the circuit shown below using 5 Volt supplies. See what happens when you reduce the size of the variable resistor.

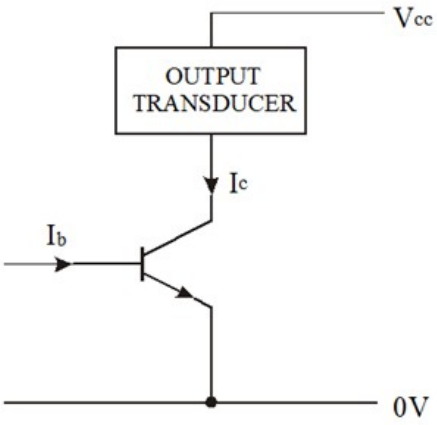


Now re-design the circuit to include a base protection resistor.

Transducer driver circuits

Output transducers can require large currents to operate them. Their resistance tends to be small.

Currents derived from input transducers, either directly, or from using a voltage divider circuit tend to be small. A transistor circuit can be used to drive the output transducer. A small current into the base of the transistor will cause a large current to flow in the collector/ emitter circuit into which the output transducer is placed (see fig 18).



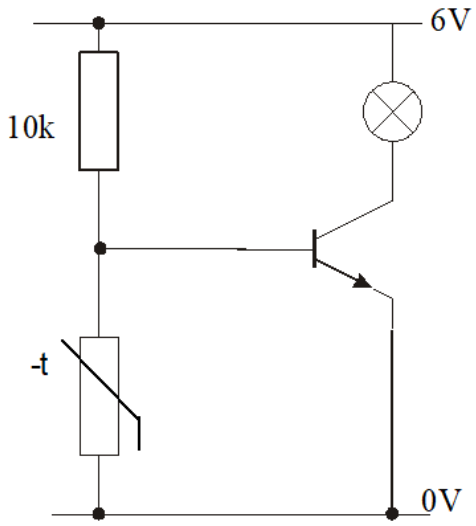
The base current is derived from applying a voltage to the base of the transistor.

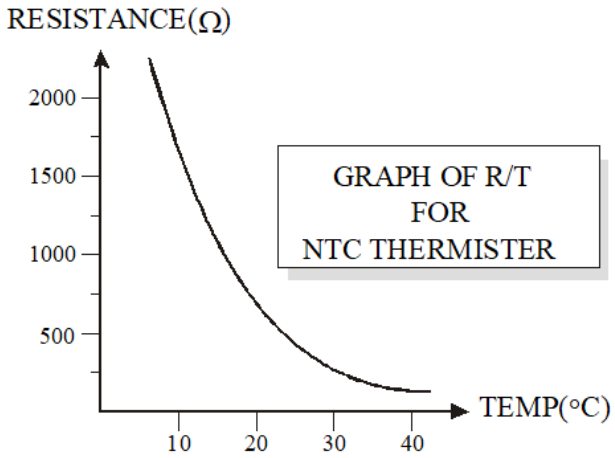
If the voltage between the base - emitter junction (V_{be}) is less than 0.6V, the transistor will not operate, no current will flow in the emitter/collector circuit and the output transducer will be off.

If V_{be} is 0.7V (or forced above 0.7V), the transistor will operate, a large current will flow in the emitter/collector circuit and the transducer will switch on.

If V_{be} lies between 0.6 and 0.7, the transistor acts in an analogue manner and this may result in the output transducer hovering around an on and off state.

28 A type 1 NTC thermistor is used in the circuit shown below to indicate if the temperature falls too low. When the bulb is on the current through it is 60 mA.





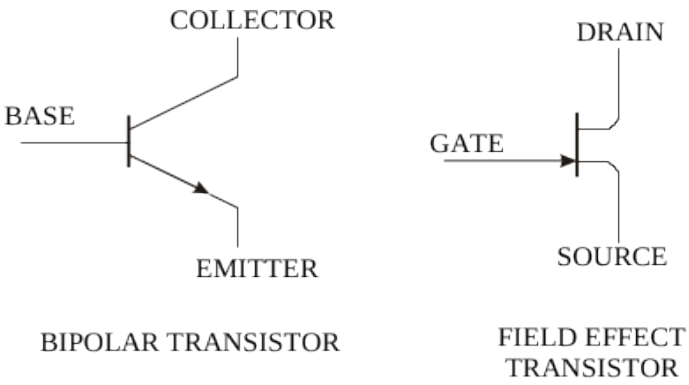
a) If h_{FE} for the transistor is 500, determine the base current required to switch on the bulb.
b) What voltage is required at the base of the transistor to ensure that the bulb indicator switches on?
c) Calculate the voltage dropped over, and hence the current through the 10k resistor.
d) Calculate the current through the thermistor and the resistance of the thermistor when the bulb is on?
e) Using the information on the graph, determine at what temperature the bulb would come on.
f) How could the circuit be altered so that the bulb would come on at a different temperature?
g) How could the circuit be altered so that the bulb would come when the temperature is too high?

N.B. The voltage between the base and emitter junction always remains at around 0.7V even if the input voltage is increased. This occurs because the base/emitter resistance decreases. This in turn causes an increase in the base current (with no increase in collector current since the transistor is already turned fully on) and as we have already seen, circuits of the type shown above could damage the transistor. It is normal practice to include a resistor in either the base or / and emitter line to limit the base current. In either case, the base - emitter voltage will be around 0.7V.

MOSFETs

Although the base current in a transistor is usually small (< 0.1 mA), some input devices (e.g. a crystal microphone) may be limited in their output. To overcome this, a Field Effect Transistor (FET) can be used.

There are many types of FETs but this course will focus on MOSFETs:
Metal Oxide Semiconductor Field Effect Transistors.



In National 5, relays were used to control higher power circuits from low power circuits, but in the real world this is not always practical. A relay can turn on and off a few times a second, but a MOSFET can switch much faster; tens of billions of times in a second. Also MOSFETs can be made much smaller, on the nanoscale. There are millions of them in your smart phone (making up the digital logic gates inside it)!

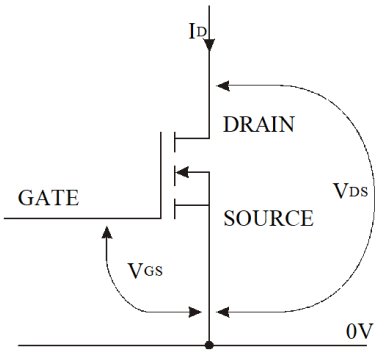
A MOSFET works in a similar way to a BJT, however a BJT is current-operated and a MOSFET is voltage-operated. Applying a voltage to the gate connection allows current to flow between the drain and source connections. This voltage operated device has a very high input resistance (unlike the BJT) and therefore requires very little current to operate it (typically 10-12 µA).

Since it operates using very little current, it is easy to destroy a FET just by the static electricity built up in your body. Touching one with your finger would completely destroy it!

FET's also have the advantage that they can be designed to drive large currents, they are therefore often used in transducer driver circuits.

Operation and Saturation

For a given MOSFET, the size of the current between the drain and source will depend on the gate voltage (V_{GS}) and the voltage between the drain and source (V_{DS}).



Like a bipolar transistor, if the gate voltage is below a certain level (the threshold value, V_T), no current will flow between the drain and source (the MOSFET will be switched off).

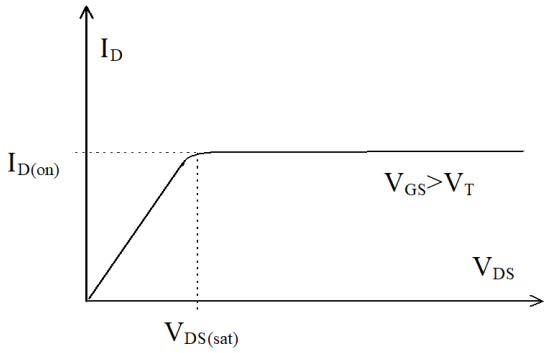
If the gate voltage is above V_T, the MOSFET will start to switch on.

Increasing the gate voltage will increase the thickness of the channel, increasing the number of charge carriers in the channel and hence increasing I_D.

For a given value of V_{GS} (above V_T), increasing V_{DS} increases the current until saturation occurs. Any further increase will cause no further increase in I_D. The MOSFET is fully on and can therefore be used as a switch.

Saturation occurs when $V_{DS} = V_{GS} - V_T$.

If V_{DS} is ≥ $V_{DS\ saturation}$, I_D is constant (for a given value of V_{GS}) (I_D is then known as $I_{D(on)}$).



When saturation occurs $I_D = I_{D(on)}$

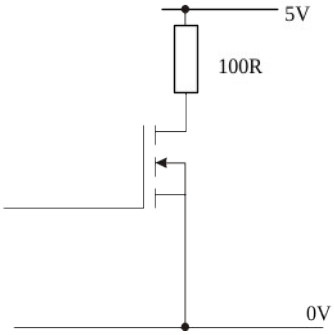
When saturation occurs, the resistance of the channel, R_{DS}, is normally low ($R_{DS(on)}$) less than 1Ω for power MOSFETs)

All other transistor rules apply.

For example, rather than $V_{CC} = V_{Out} + V_{Load}$ we can now say $V_{CC} = V_{DS} + V_{Load}$

Worked Example

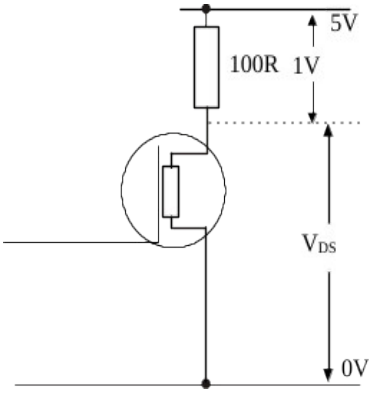
The threshold gate voltage for the MOSFET shown below is 2V. Calculate the gate voltage required to ensure that a saturation current of 10 mA flows through the load resistor.



Step 1

The drain - source channel acts as a series resistor with the 100R, since the current is the same in a series circuit, the voltage over the 100R can be calculated using Ohm's law

V = IR = 10 mA x 100 = 1V



Step 2

Using Kirchoff's 2nd law, the voltage over the channel + the voltage over the load resistor = supply voltage hence

V_{DS} = 5 - 1 = 4V

Step 3

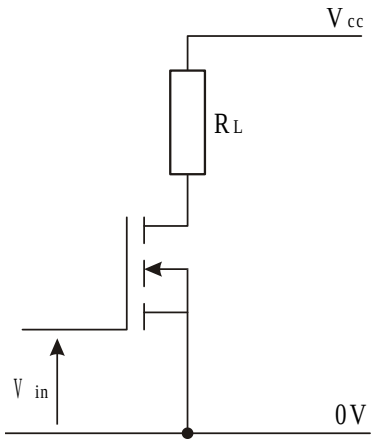
For saturation to occur,

$V_{DS} = V_{GS}-V_T$

$V_{GS} = V_{DS} + V_T$

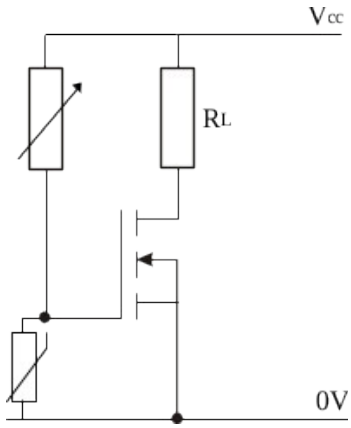
$V_{GS} = 4 + 2 =$ **6 Volts**.

MOSFET's can be designed to handle **very high drain currents**, this means that they can be used to drive high current output transducers drivers without the need for relay switching circuits (unlike the bipolar transistor).



The load resistor could be any output transducer, bulb, motor, relay etc. Since MOSFET's are particularly sensitive to high voltages, care must be taken to include a reverse biased diode over transducers that may cause a back emf when switched off.

A variable resistor can be used in a voltage divider circuit and adjusted to ensure that the input voltage to the gate = V_T

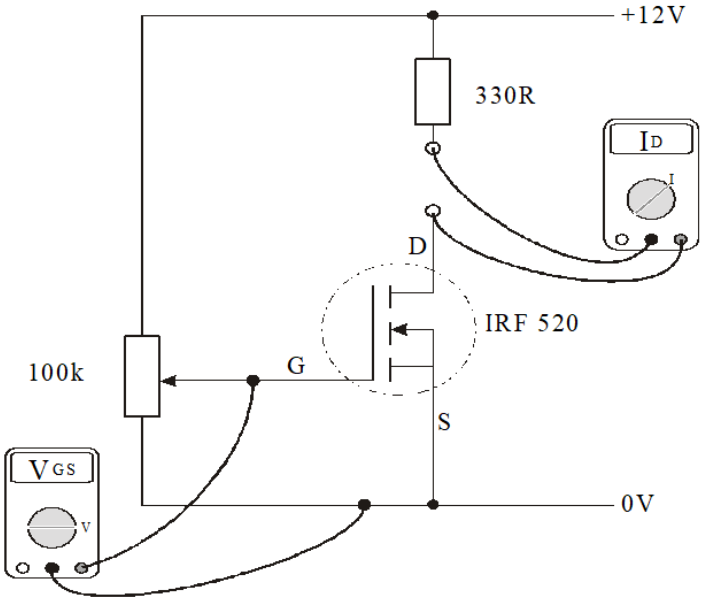


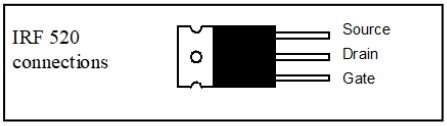
29 Create a table comparing MOSFETs and BJT Transistors. Be sure to include all advantages and disadvantages.

Practical/ Simulation questions

MOSFET characteristics - relationship between Drain current (I_D) and Gate voltage (V_{GS})

1. Construct the circuit shown below using multi-meters to measure the drain current and the gate voltage.



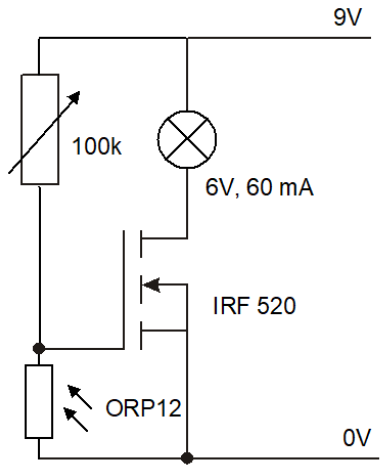


V_{GS} (V)	I_D (mA)

Adjusting the potentiometer will allow you to change V_{GS}.

a) Construct a table for recording your results (as shown to the side).
b) Measure I_D for various values of V_{GS}.
c) Find the threshold voltage for this MOSFET
d) Plot a graph of I_D against V_{GS} and calculate g_m for this MOSFET.

2. Construct the circuit shown below.

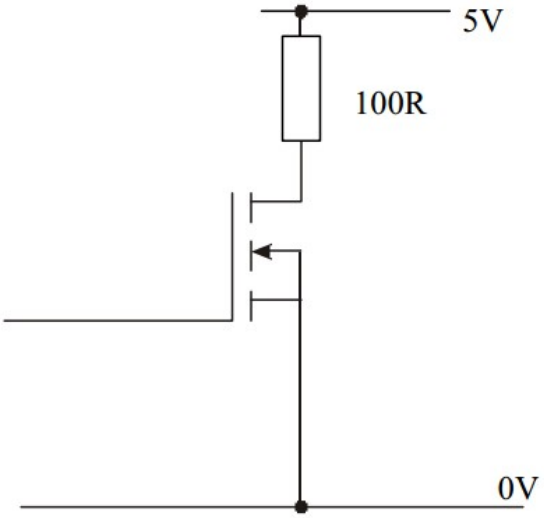


Adjust the 100k variable resistor until the light just goes off. (This means that $V_{GS} < V_T$)

Covering up the LDR will increase V_{GS} and the light should come on.

3

a) Explain the term threshold voltage, as applied to MOSFETs
b) The threshold gate voltage for the MOSFET shown below is 2V. Calculate the gate voltage required to ensure that a saturation current of 10 mA flows through the load resistor.



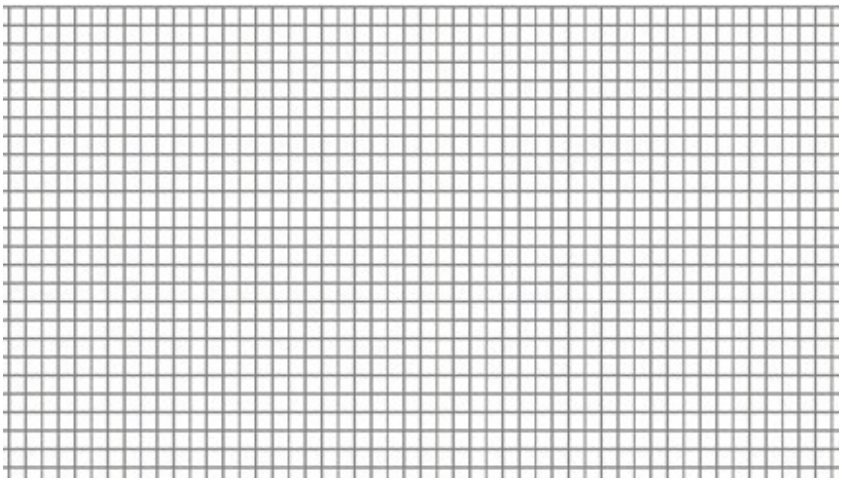
31

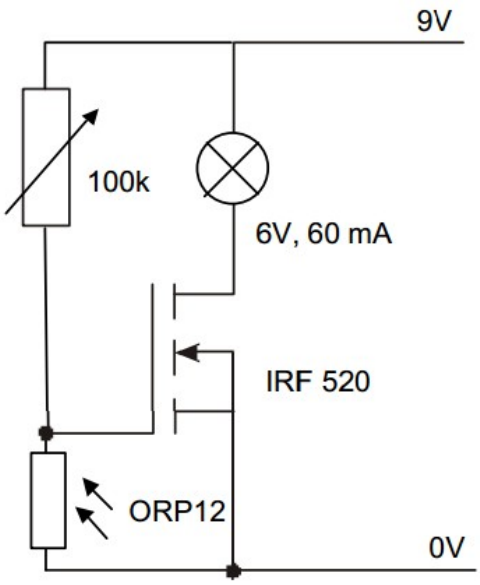
A car is fitted with a two-speed heater fan. The power to the heater-fan motor is controlled by a single pin from a microcontroller. Assuming no losses in the output driver, the heater-fan motor receives either 100% or 60% of the full power available.

A MOSFET output-driver circuit provides current to the heater-fan motor. Draw the output-driver circuit showing connections to the motor, the power-supply rails, and to the output pin of the microcontroller. (4 marks)

Practical question

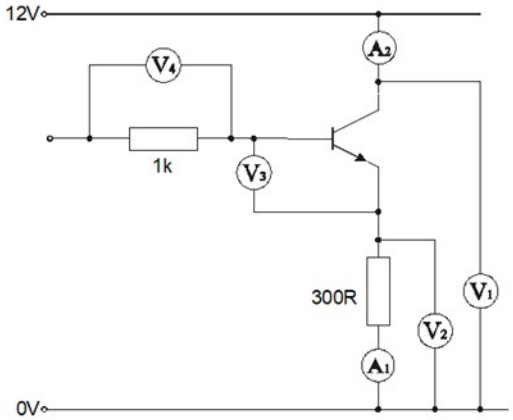
Create the circuit below using circuit simulation software. Plot a graph of I_D against V_{GS} for this MOSFET. What do you notice?





32 The following electronic system is set up for a test with various ammeters and voltmeters connected as shown in Figure Q1. In the condition shown, the transistor is fully saturated with a base current of 500 μA.

Write down the readings which you would expect to see on each of the four voltmeters (V_1 - V_4) and the two ammeters (A_1 - A_2).



33 The circuit shown in Figure Q4 is designed to switch on a greenhouse heater when the temperature in the greenhouse falls to 15 °C.

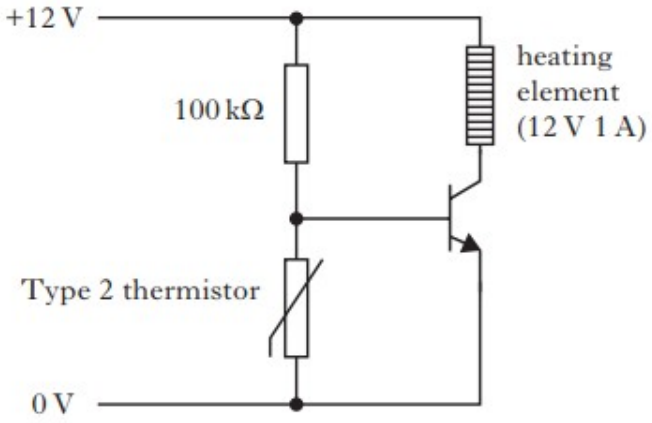


Figure Q4

(*a*) Calculate, for a temperature of 15 °C with the transistor switched on and saturated:

(i) the current flowing through the 100 kΩ resistor; **2**

(ii) the current flowing through the thermistor; **3**

(iii) the current flowing into the base of the transistor. **1**

The heating element is rated at 12 V, 1 A.

(*b*) Calculate the minimum current gain, h_{FE}, required for the heating element to fully switch on at 15 °C. **2**

A signal from a logic circuit controls the operation of a 12 V lamp by means of the transistor driver shown in Figure Q4.

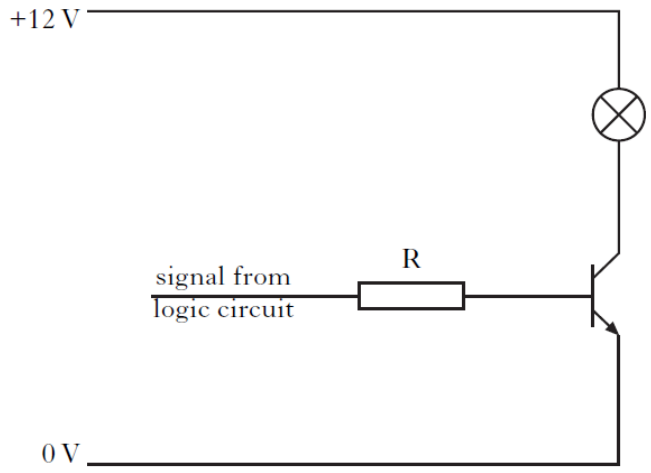


Figure Q4

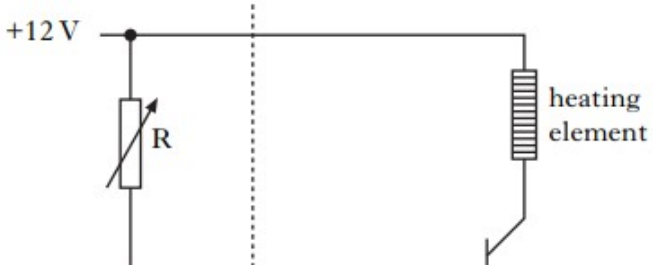
When the signal from the logic circuit is high, the voltage is 12 V and the current is 3 mA.

(*a*) Calculate the value of resistor R. **1**

The lamp is rated at 20 W at its normal operating temperature. However, when the lamp is cold it draws a much higher current and so the transistor must be capable of handling 5 times the normal operating current of the lamp.

(*b*) Calculate the minimum required current gain of the transistor. **2**

35 The circuit shown in Figure Q2 represents a system for controlling a heater in a greenhouse.



36

Marks

Figure Q2 shows a circuit for controlling a motor connected to a fan.

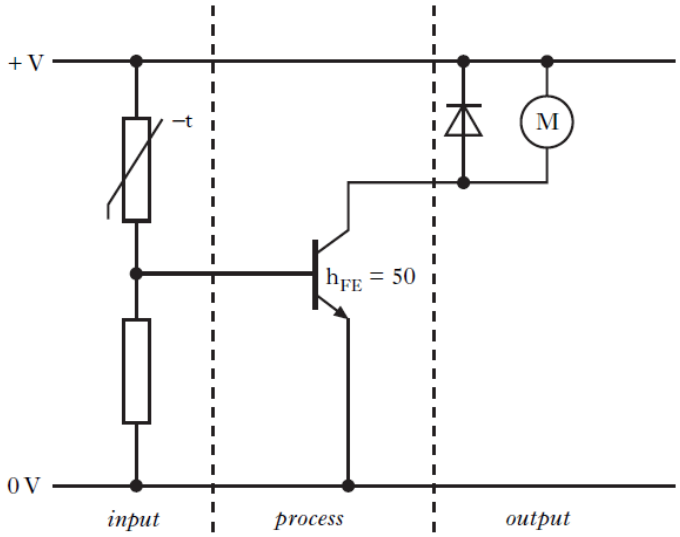


Figure Q2

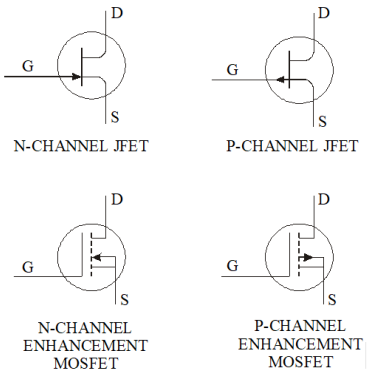
(*a*) State the name of the protective component in the circuit and explain its function. **1**

(*b*) Explain the function of the circuit by describing how the *input* sub-system responds to increasing temperature, and how the resulting signal from the *input* sub-system affects the *process* and *output* sub-systems. **3**

Additional MOSET information

This page has some additional reading on MOSFETs, however it is not essential for our Higher course. If you want to know how MOSFETs work, read on, else skip this page.

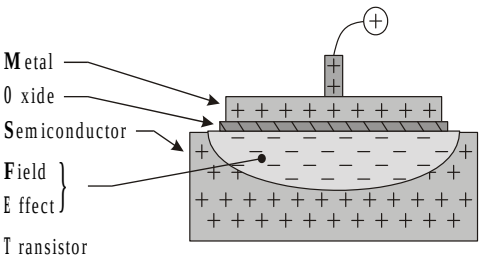
While there are two different types of FET's are available, JFET (Junction Field Effect Transistor); and MOSFET (Metal Oxide Semiconductor Field Effect Transistor), all FET's can be N-channel or P-channel and MOSFETs can be "depletion type" or "enhancement type".



The simplicity in construction of the MOSFET means that it occupies very little space. It can also be designed to be used as a resistor or capacitor. Because of its small size, many thousands of MOSFET's can easily be incorporated into a single integrated circuit. The high input resistance means extremely low power consumption compared with bipolar transistors. All these factors mean that MOS technology is widely used within the electronics industry today. Enhancement-type MOSFET's can be used in a similar way to bipolar transistors. N-channel enhancement MOSFET's allow a current to flow between Drain and Source when the Gate is made Positive (similar to an NPN transistor). P-channel enhancement MOSFET's allow a current to flow between Drain and Source when the Gate is made Positive (similar to an PNP transistor).

N-channel enhancement MOSFET - theory

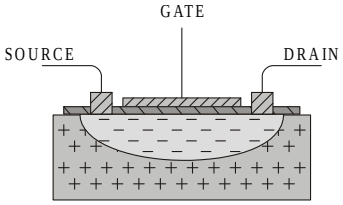
The MOSFET transistor is constructed on a piece of p-type Semiconductor, an Oxide insulating layer separates this from the Metal 'Gate' contact.



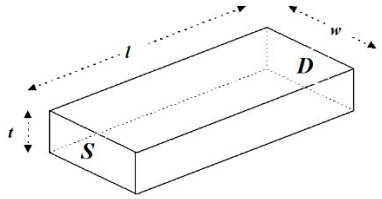
Making the metal contact positive will produce a Field, repelling the positive majority carriers leaving negatives behind (to produce an N-channel).

If the field produced by the Gate voltage is large enough, a channel is produced connecting the Drain and Source. When a voltage is applied between the Drain and Source, current will flow along the channel.

(In practice, the Drain and Source metal plates are connected to n-type semiconductor within the p-type substrate).



The channel acts as a resistance between the Source and Drain. The size of which depends on:

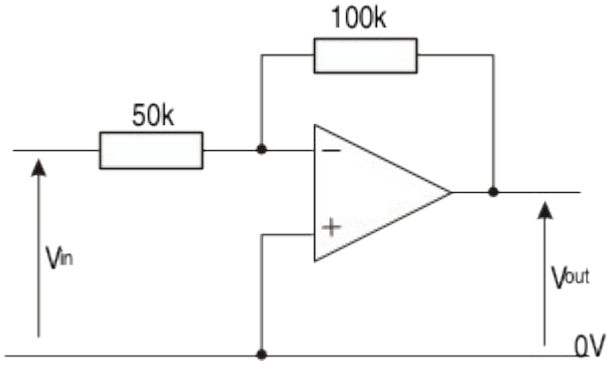


- the length (*l*) of the channel between D and S
- the width (*w*) of the channel
- the thickness (*t*) of the channel

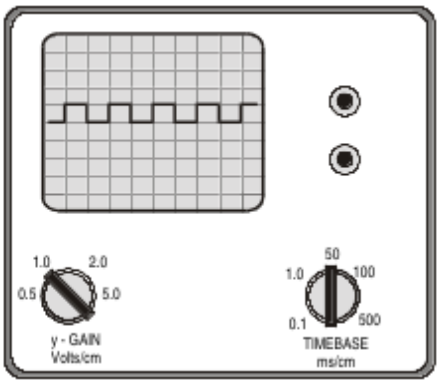
L and *W* are determined during manufacture of the MOSFET, *t* is adjusted by altering the gate voltage.

Exam Style Questions

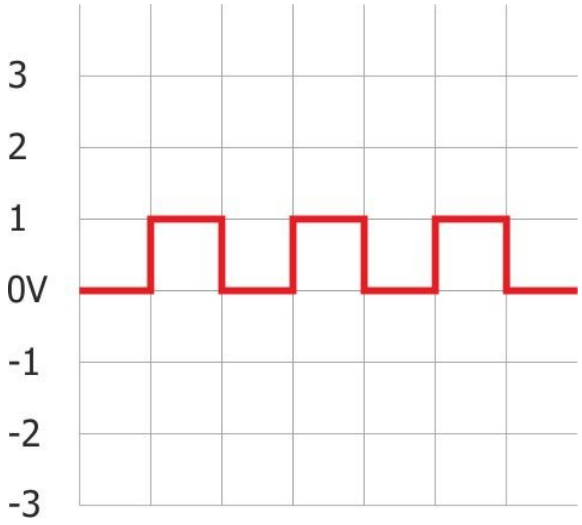
1. An op-amp is connected as shown in below.



The input to the circuit is monitored by a single beam oscilloscope. The diagram below shows the screen display and the control settings.



a) Determine the frequency of the pulses at the input. (each box on the screen represents 1cm).
b) Assuming the oscilloscope controls are not adjusted, redraw the trace you would expect to see if the oscilloscope was connected to the output of the circuit. (The input trace is shown in red).



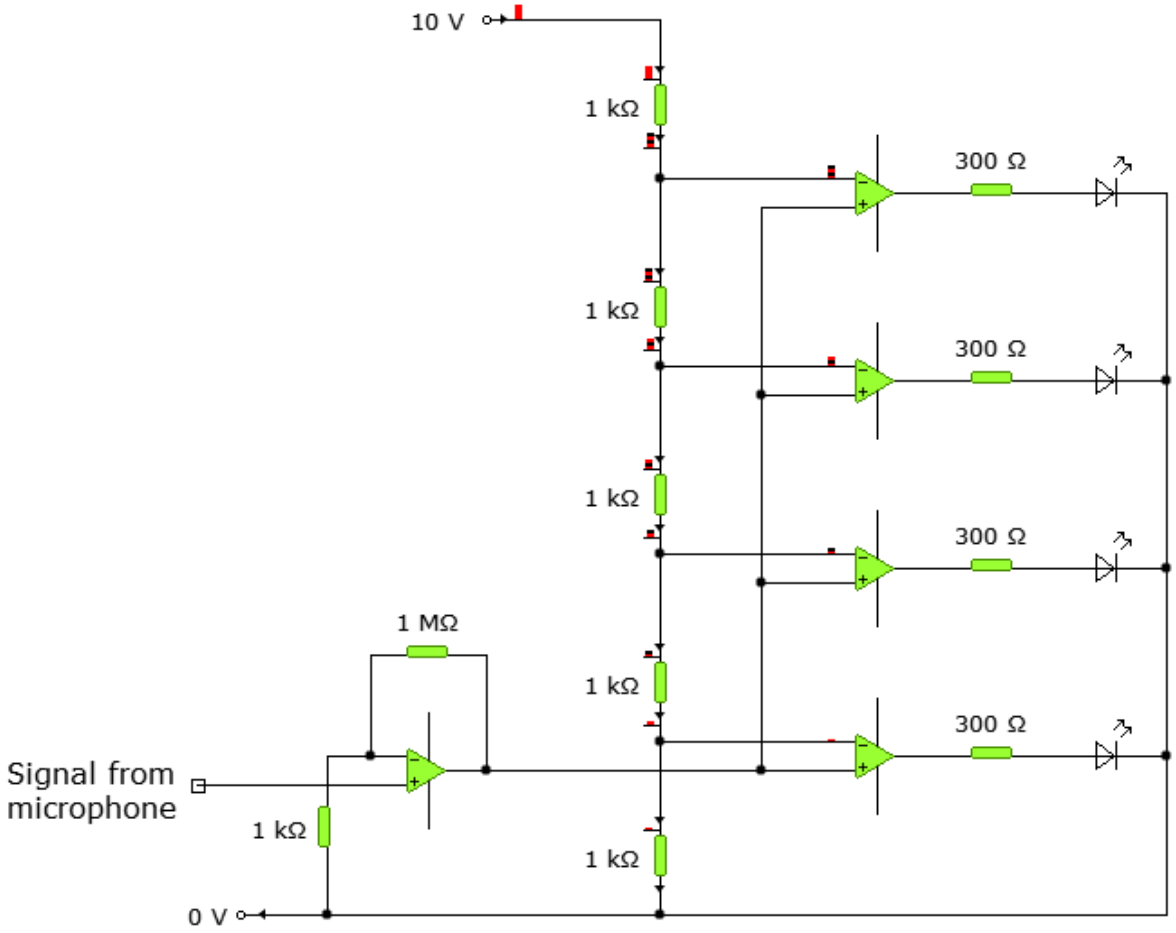
c) The same components are now used to wire the op-amp in the non-inverting configuration. Re-draw the new circuit diagram and calculate the output voltage you would expect.

2. When a tacho generator is rotated, it produces a voltage proportional to its angular velocity. The tacho is to be used to monitor the speed of coolant flowing along a pipe. If the speed is below a recommended level then an alarm should come on, if the speed is above this level then a green light should come on.

   Design and simulate a circuit that will meet the following criteria;

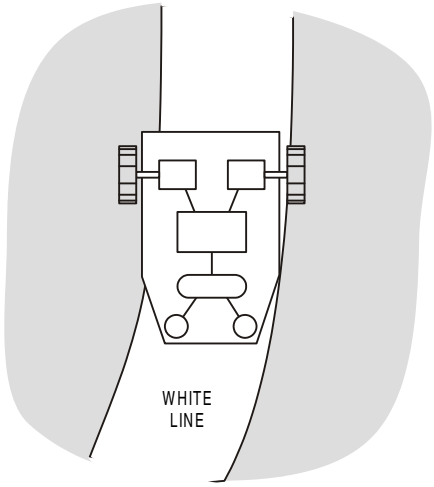
   - A user must be able to set a reference voltage for the system
   - When the voltage from the tachogenerator (simulate this with a variable voltage supply) is above the reference level a green light should come on.
   - When the voltage is below the reference a warning buzzer must sound.
   - The buzzer and light must be powered by a separate supply.

3. Part of a disco audio system consists of the microphone connected to a circuit containing two op-amps configurations as shown below. (Assume each op-amp is supplied with suitable voltage).



a) Explain clearly how the circuit operates.
b) Why are two op-amp configurations used in this circuit?

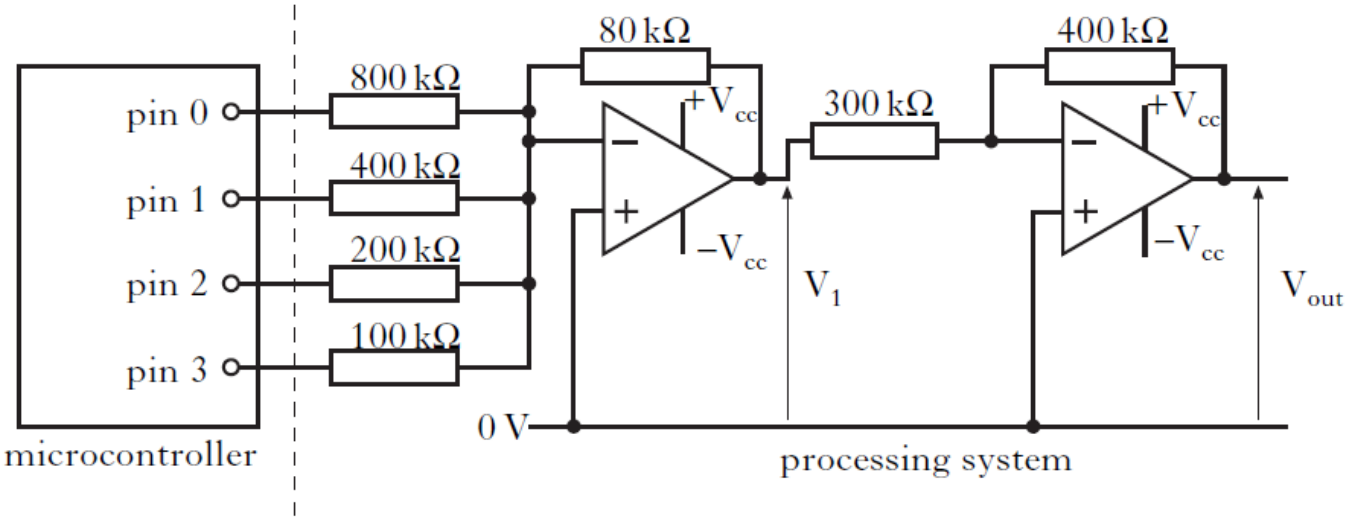
4. A robot is designed to follow a white line painted on the floor. The robot moves by means of suitably geared motors which can be switched on and off independently.



The left sensor controls the right motor (and vice versa). When a sensor detects a white line, the motor is switched on.

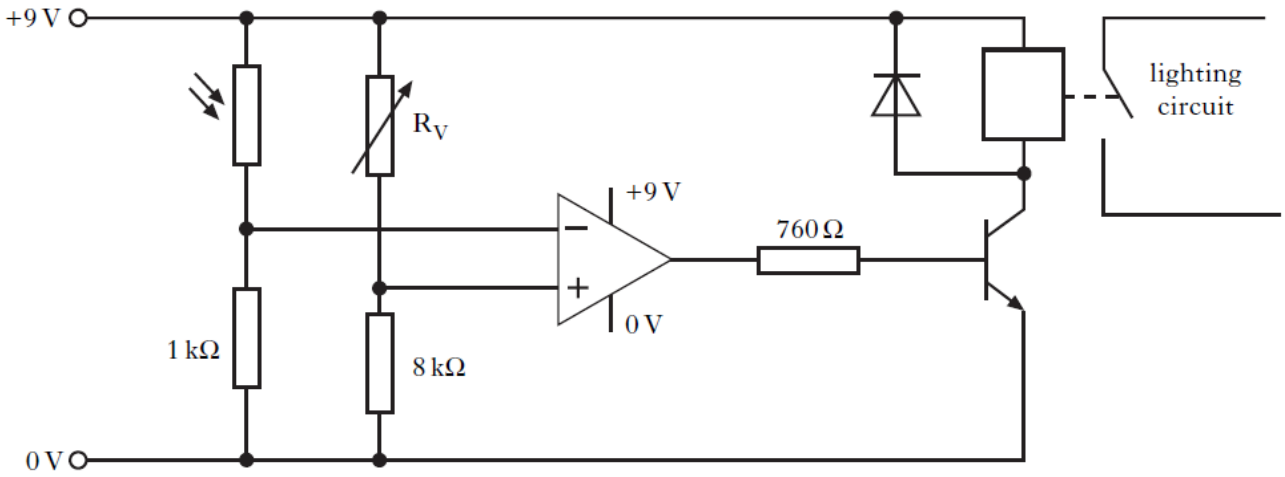
a) Suggest a suitable detector that could be used to detect the white line.
b) Design a suitable circuit that could be used to control one of the motors.

5. The diagram below show a processing system based on op-amps which process digital signals from a microcontroller and converts them to analogue.



a) Calculate the value of V_1 and V_{out} when pins 0 and 2 are switched on. (Each pin will produce a 5V output when switched on
b) Calculate the maximum value of V_{out}.

6. An energy –saving system is installed in a new office block. If the natural light level in an office is high enough, the lights turn off automatically. The control circuit is shown below.



a) Calculate the value of R_v so that the light switches off at 650lux.

b) Calculate the base current when the op-amp output is high.

Thermistor graph

Light Dependent Resistor (LDR) graph for an ORP12 LDR